Computers and Chemical Engineering 163 (2022) 107810

Contents lists available at ScienceDirect

Computers and Chemical Engineering

journal homepage: www.elsevier.com/locate/compchemeng

Techno-economic and environmental analysis of methanol and dimethyl ether production from syngas in a kraft pulp process

Meire Ellen Gorete Ribeiro Domingos[a,c,*], Daniel Flórez-Orrego[b,c,d], Moisés Teles dos Santos[a], Silvio de Oliveira[b], François Maréchal[c]

[a] Department of Chemical Engineering, Polytechnic School, University of Sao Paulo, Sao Paulo, Brazil
[b] Department of Mechanical Engineering, Polytechnic School, University of Sao Paulo, Sao Paulo, Brazil
[c] Industrial Process and Energy Systems Engineering (IPESE), École Polytechnique Fédérale de Lausanne, Sion, Switzerland
[d] Department of Processes and Energy, BioFR Group, Faculty of Minas, National University of Colombia, Medellin, Colombia

ARTICLE INFO

Article history:
Received 1 November 2021
Revised 12 March 2022
Accepted 17 April 2022
Available online 28 April 2022

Keywords:
Black liquor
Kraft process
Gasification
Methanol
Dimethyl ether
Economic analysis
Environmental analysis

ABSTRACT

The conventional configuration of a pulp mill, in which black liquor is concentrated and burnt, is compared to the upgrading gasification routes for methanol (MeOH) or dimethyl ether (DME) co-production. To this end, various exergy-based and environmental impact indicators are assessed in the light of different utility supply scenarios. The combined exergy and energy integration analysis are used to identify potential improvements related to the decarbonization and mitigation of the process irreversibility. As a result, the exergy efficiencies of the conventional scenario and the integrated plants average 40% and 45%, respectively, whereas the overall CO_2 emission balances vary from 1.97 to -0.07 t_{CO2}/t_{Pulp}, respectively. Additionally, an incremental economic analysis that envisages future carbon taxation scenarios suggests that only MeOH or DME co-production routes with partial electricity import may economically outperform the conventional kraft pulp mill for moderate carbon taxations. These results highlight the relevance of the electricity import from the Brazilian mix for pushing upwards the share of renewable energy resources in the production of traditionally fossil-based fuels and chemicals.

© 2022 The Author(s). Published by Elsevier Ltd.
This is an open access article under the CC BY-NC-ND license
(http://creativecommons.org/licenses/by-nc-nd/4.0/)

1. Introduction

The concerns related to the security of energy and other important commodities, along with the awareness about the effect of the climate change, have boosted the interest in biomass-based energy resources as alternatives to substitute the fossil feedstock (Nakashima et al., 2019). In this regard, the pulp and paper industry can play an important role, as it is a well stablished biomass-based activity (Domingos et al., 2019). However, pulp and paper sector is a very energy-intensive industry (Ericsson and Nilsson, 2018), which renders it responsible for about 2.3% of the total yearly CO_2 emission of the industrial sector worldwide (IEA, 2020). In this way, strategies such as the electrification of some of its processes for heating purposes could promote the decarbonization of this industry if the electricity comes from a low-carbon source (Ericsson and Nilsson, 2018).

A potential pathway for long-term sustainable growth of the pulp and paper industry that can improve its efficiency and profitability is the integration of biorefineries into the existing facilities (Mongkhonsiri et al., 2018). The black liquor (BL) is a byproduct of the kraft pulping process, which can be an important feedstock for the production of value-added products, considering that it contains more than half of the original energy content in the total woody biomass. In this way, the syngas produced from the gasification process of BL can be used either to generate electricity or produce chemicals and biofuels, such as methanol (MeOH) and dimethyl ether (DME). Thus, the products portfolio can be expanded, matching the future concept of the biorefineries (Domingos et al., 2021).

Abbreviations: BFW, Boiler feedwater; BL, Black liquor; BLG, Black liquor gasification; CAPEX, Capital expenditure; CW, Cooling water; DC, Distillation column; DEPG, Diethyl propylene glycol; DME, Dimethyl ether; FEHE, Feed-effluent heat-exchanger; INPV, Incremental net present value; H, Enthalpy flow rate (kW); HTS, High temperature shift; LHT, Liquor heat treatment; LHV, Lower heating value (kJ/kg); LHHW, Langmuir-Hinshelwood Hougen-Watson; LTS, Low temperature shift; MeOH, Methanol; MER, Minimum energy requirement; Mt, Million tons; MVR, Mechanical vapor recompression; OPEX, Operating costs; WGS, Water gas shift.

* Corresponding author at: Department of Chemical Engineering, Polytechnic School, University of Sao Paulo, Sao Paulo, Brazil.
E-mail address: meireellengorete@usp.br (M.E.G. Ribeiro Domingos).

https://doi.org/10.1016/j.compchemeng.2022.107810
0098-1354/© 2022 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license
(http://creativecommons.org/licenses/by-nc-nd/4.0/)

Nomenclature

Latin symbols

b	Specific chemical exergy (kJ/kg)
B	Chemical exergy (kW)
C	Carbon mass fraction (%)
C	Market costs and selling prices for feedstock and products (€ per kWh or kg)
C_0, C_1	Cost at the reference scale (€)
E	Activation energy (kJ/mol)
Exp	Expenses (€)
f	Unit load optimization factor (-)
h	Specific enthalpy (kJ/kg)
h_0	Specific enthalpy at standard conditions (kJ/kg)
h_{lv}	Enthalpy of evaporation of water at standard conditions (kJ/kg)
H	Hydrogen mass fraction (%)
HHV	Higher heating value (kJ/kg)
i	Interest rate (%)
k_0	Rate constant (kmol/(m^3.s.Pa))
M	Molecular mass (kg/kmol)
MC_{eq}	Equilibrium moisture content (%)
m	Mass flow rate (kg/h)
N	Number of utility units, temperature intervals, yearly periods, nitrogen mass fraction (%)
n_{mix}	Total molar flow of the mixture (kmol/h)
O	Oxygen mass fraction (%)
P	Partial pressure (bar)
Q	Heat (kW)
q	Cooling or heating duty from the utility systems (kW)
R	Universal gas constant, cascaded heat rate (kW)
Rev	Revenues (€)
r	Reaction rate kmol/(m^3.s), scale exponent, indirect emissions factor (g_{CO2}/kJ)
SN	Stoichiometric number (-)
s	Specific entropy (kJ/kg-K)
s_0	Specific entropy at standard conditions (kJ/kg-K)
S_0	Reference scale capacity (various)
S_1	Actual scale capacity (various)
T	Temperature (°C)
$\vec{v}$	Velocity (vector) (m/s)
W	Power (kW)
x_i	Mol fraction (%)
y	binary (existence) optimization factor
γ_i	Activity coefficient (-)
z	Elevation (scalar) of the system (m)

Greek symbols

λ	Renewability index (-)
φ	Ratio of the chemical exergy to the lower heating value
η	Exergy efficiency (%)
ω	Utility or process unit

Subscripts

AD	Air dried
BL	Black liquor
Dest.	Destroyed
DME	Dimethyl ether
exp.	Exported
imp.	Imported
MEOH	Methanol
r	Interval of temperature

Superscripts

CH	Chemical exergy
K	Kinetic
P	Potential
PH	Physical

Methanol is a very important chemical in the energy sector, where it can be used directly as fuel or blended with gasoline. Also, it can be used as an intermediate molecule for various chemical syntheses, such as olefins, amines, acetic acid, dimethyl ether and formaldehyde (Leonzio, 2018). Dimethyl ether has remarkable properties that make it useful in different fields such as power generation, household, transport, as refrigerant fluid and as feedstock to synthesize other products. DME is considered as non-toxic, non-corrosive, colorless and environmentally friendly compound. It also has the advantage to be easily transportable, to have a high cetane number and similar properties to the liquefied petroleum gas (Mevawala et al., 2017).

Traditionally, the methanol and dimethyl ether industrial processes are based on fossil sources, such as coal and natural gas, which present some disadvantages, such as high energy and water consumption and high level of pollutant emissions, not to mention their nonrenewability and future depletion (Yang et al., 2018). Thus, alternative routes have been investigated aiming to decarbonize their production supply chains, such as the biomass thermochemical conversion via gasification (Yang et al., 2018; Carvalho et al., 2018).

The methanol and dimethyl ether production via black liquor gasification has been studied by several authors (Ekbom et al., 2003; Häggström et al., 2012; Fornell et al., 2013; Naqvi et al., 2010b; Pettersson and Harvey, 2012; Carvalho et al., 2018). Häggström et al. (2012) successfully synthesized methanol from syngas generated via black liquor gasification. In their work, the syngas passes through a catalytic reactor at 25 bar, which yields 0.16–0.19 $g_{methanol}/(g_{catalyst}.h)$. This experiment was an important step towards demonstrating the feasibility of methanol synthesis from a bench scale for subsequent scale-up processes.

When a conventional kraft pulp mill is integrated to biofuels production, the existing cogeneration unit falls short to supply the increased utilities requirements. This energy deficit should be fulfilled by importing additional biomass to compensate for the energy withdrawn from the black liquor (Naqvi et al., 2010a). Ekbom et al. (2003) studied the technical and economic feasibility of BL gasification for motor fuels production (MeOH and DME). The authors evaluated the marginal energy efficiency of methanol and dimethyl ether production by considering additional biomass consumption. The biomass-to-fuel conversion efficiency was 66% and 67%, for methanol and dimethyl ether, respectively. In both cases, the energy efficiency is enhanced due to the increased energy flows of both biomass and motor biofuels produced.

Carvalho et al. (2018) compared the conventional BL combustion to an integrated methanol production via BL gasification and other blended feedstock. The integrated system resulted in a higher efficiency (60%) compared to the conventional combustion of BL to generate steam and electricity (below 40%). The pure BL gasification case also presented higher energy efficiency compared to the co-gasification scenarios, partly due to the increased energy demands of these latter cases (Carvalho et al., 2018).

Fornell et al. (2013) performed a techno-economic analysis of a biorefinery built upon the integration of a kraft pulp-mill and ethanol and dimethyl ether production plants. The pulp is sent to an ethanol production process, while the BL is gasified and refined to produce dimethyl ether. The results of this study indicated that the process can be self-sufficient in terms of hot util-

ity demand (i.e. fresh steam). However, a deficit of electricity is expected.

Pettersson and Harvey (2012) compared the BL gasification and downstream production of DME or electricity to other pulping biorefinery concepts, considering different energy market scenarios. The authors observed that BL gasification with DME production has the best economic performance for both pulp mills and integrated pulp and paper mills. The carbon capture and storage was also considered in this study, and it pointed to improved process profitability.

Despite the relevance of the previous works, they only considered thermodynamic or environmental indicators. A holistic approach that considers both performances, along with the economic implications of the feedstock and fuels price variation over the plant lifespan, has not been addressed. Thus, in this work, the scenario of the conventional utilization of black liquor in a kraft pulp mill (i.e. concentration and combustion) is compared to the BL upgrading gasification process for methanol (MeOH) and dimethyl ether (DME) production. Those fuels are reported as potential low-carbon fuels that may help decarbonizing the transportation sector (Fernández-Dacosta et al., 2019). The exergy destruction and the cumulative CO_2 emissions are considered as environmental criteria for assessing the sustainability of the energy conversion systems. The extended exergy method takes into account the inefficiencies of the different upstream supply chains, which provides a more accurate estimation of the environmental impacts. Finally, an incremental techno-economic assessment helps elucidating the effect of the uncertainty of the feedstock costs and the carbon taxes on the financial feasibility of the proposed processes.

2. Methods

This section describes the processes modeling and simulation and the tools used for the calculation of the stream properties. The environmental, economic and thermodynamic performance indicators, suitable for carrying out the incremental comparative analysis, are also defined. Lastly, the optimization method, used to minimize the energy requirements, while maximizing the waste heat recovery, is also discussed.

2.1. Process modeling and simulation

Fig. 1 shows the flowsheet of the integrated processes: the pulp and methanol or pulp and dimethyl ether production plants. Each unit reported in the flowsheet will be described in the following sections. The Aspen Plus® v8.8 software is used to evaluate the mass, energy and exergy balances for all units, considering the Peng-Robinson equation of state with Boston-Mathias modifications. The perturbed-chain statistical associating fluid theory (PC-SAFT) correlation is used to model the physical absorption of CO_2 with dimethyl ethers of polyethylene glycol (DEPG) (Adams II et al., 2014). Finally, the NRTL-RK property method (nonrandom two liquid activity-coefficient model with Redlich−Kwong equation of state) is used for modeling the distillation section of MeOH and DME, since it is suitable for describing the phases present in those systems (Kiss et al., 2016).

2.1.1. Kraft pulp mill

First, the logs are debarked, cut into wood chips and classified (Fig. 1, kraft pulp mill section). Approximately 10% of the initial wood mass is lost in these processes (Foelkel, 2017). The wood chips follow to the pulping process, where they are impregnated with the cooking liquor (white liquor, approx. 60% NaOH, 25% Na_2S and 15% Na_2CO_3), and fed to the digesters. The pulp yield varies between 45 and 55% of the initial wood mass, depending of the process conditions and the biomass quality (Moraes, 2014; Rural Technology Initiative, 2016). In order to recover the cooking liquor (as black liquor) and to reduce the carryover of organic components to the next step, the pulp coming from the digester is washed. The black liquor generated consists of about half of the organic material that was initially in the wood chips and almost all of the inorganic chemicals that were used for the pulping process. The literature reports that the amount of dry solids in the black liquor per unit of pulp produced ranges from 0.86 up to 1.7 t_{BL}/t_{ADpulp} (IEA, 2007; Bachmann, 2009). Next, the pulp is bleached with hydrogen peroxide and ozone in a sequence of acidic and alkaline stages, with interwashing (Naqvi et al., 2010a). Finally, the bleached pulp is dried until it achieves 90% solids content, and prepared to be commercialized.

In this work, it was considered a pulp yield of 46.51% with respect to the total amount of digested biomass (Foelkel, 2017), with a respective production of 1.44 t_{BL}/t_{pulp}. The power and steam demands are adapted from (Moraes, 2014; Ferreira and Balestieri, 2015) to produce about 880 air dried (AD) tons of pulp per day.

2.1.2. Black liquor gasification unit

The weak BL coming from the kraft mill must be concentrated before enters the gasifier. To this end, a mechanical vapor recompression (MVR) system is used, which substitutes the large steam requirement in the BL concentration by a demand of electrical energy. It results in very low operating costs and higher efficiency, compared to a multiple effect evaporator operation (Darmawan et al., 2017a). Part of the moisture extracted is recycled back to the evaporator to improve the heat transfer performance, while the other part is purged and subsequently superheated and recompressed, aiming to supply the evaporation effect (Darmawan et al., 2017a 2017b). The BL leaves the system with about 85% wt. solids. Eq. (1) is used to estimate the evaporation temperature (T, in °C), based on the equilibrium moisture content, MC_{eq} (Darmawan et al., 2017a 2017b).

$$MC_{eq}(wt\%wet\ basis) = 54.678e^{-0.46T} \quad (1)$$

Since the viscosity of the BL is strongly influenced by the BL solid content, a thermal pre-treatment method, called liquor heat treatment (LHT), has been proposed (Vakkilainen, 2006). It consists of raising the temperature of the BL to about 180–190 °C, which allows the highly concentrated BL to be in a pumpable form (Vakkilainen, 2006; UNIDO, 1996).

Next, the concentrated BL goes to a pressurized entrained-flow, high temperature black liquor gasification process (PEHT-BLG), wherein drying, pyrolysis, gasification and combustion process are considered. The gasification pressure and temperature are set as 30 bar and 1000 °C (Huet, 2015), respectively. Moreover, there is a quench zone in this system that allows to recover the chemicals in the form of green liquor, which is recycled back to the kraft mill. A fraction of the char is combusted in order to supply the energy demanded by the endothermic drying, pyrolysis and reduction reactions. The composition of the materials considered in this work are reported in Table 1.

2.1.3. Syngas treatment and gas purification units

The syngas follows to the shift reactors, wherein an additional amount of hydrogen is produced at the expense of the CO and water content in the gas. The exothermic water gas shift (WGS) reaction is equilibrium-limited, but kinetically favored at higher temperatures, which results in an incomplete conversion of carbon monoxide. Thus, the WGS reaction is preferably achieved in two shift reactors, namely a high temperature shift reactor (HTS) and a low temperature shift reactor (LTS). After the mixture is partially converted in the HTS reactor(∼350 °C), the syngas is intercooled before is fed to the LTS reactor (∼220 °C). The intercooling of the

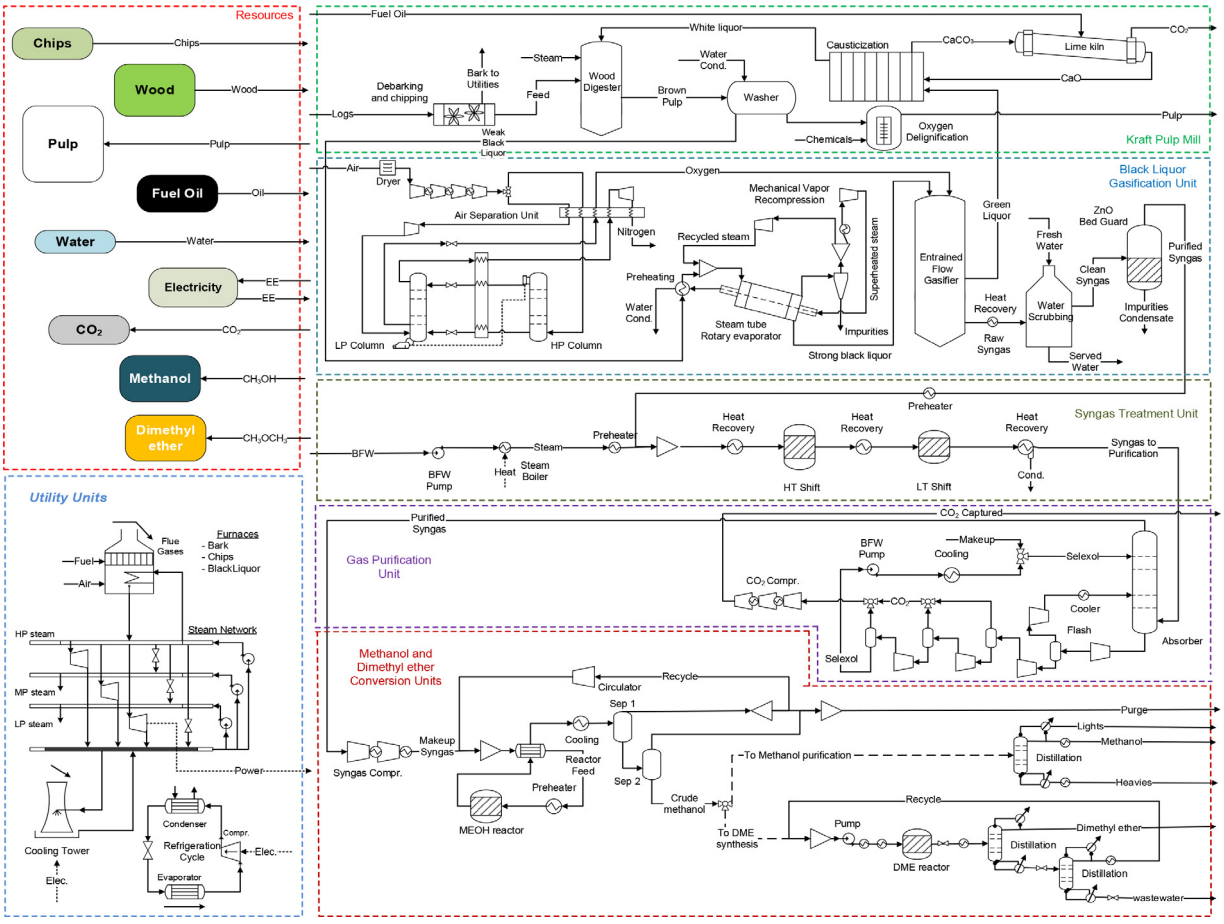


Fig. 1. Integrated flowsheet of kraft pulp mill and the methanol or dimethyl ether production plants.

Table 1
Ultimate and proximate analyses of wood, bark, black liquor, oil and pulp.

	Wood	Bark	Black liquor	Oil	Pulp
Ultimate analysis					
Carbon	49.90	48.58	29.86	85.10	44.44
Hydrogen	6.14	5.52	3.27	10.90	6.17
Oxygen	42.85	41.19	29.05	–	49.39
nitrogen	0.27	0.36	0.10	–	–
sulfur	0.00	0.03	4.09	4.00	–
Chlorine	–	0.09	0.90	–	–
Proximate analysis					
Ash	0.83	4.23	32.73	–	–
Fixed Carbon	–19.00	–15.00	10.21	–	–
Volatiles	–80.00	–80.00	57.06	–	–
Moisture	40	50	85	2	10
Refs.*	Dellavedova et al. (2012); Pio et al. (2017); Boloy et al. (2011); Macedo (2006)	Gavrilescu (2008); Theis et al. (2006)	Harris et al. (2008)	Global Combustion Systems (2019)	Demirbas (2006)

*The reported values are an average of the cited literature.

HTS and LTS reactors, as well as the condensation of the excess steam preheat the boiler feedwater (BFW), whereas further cooling is provided by using cooling water (CW). The H_2:CO molar ratio of 2, required to enter the methanol loop, is achieved by varying the amount of water in the WGS.

Next, the cold syngas produced (30 °C) is sent to the Gas Purification Unit. This system consists of a high pressure absorber (30 bar), in which a mixture of dimethyl ethers of polyethylene glycol (DEPG) contacts the syngas, producing a CO_2-rich bottoms solution. As for the methanol and dimethyl ether synthesis, the syngas composition must be adjusted considering the stoichiometric number (SN), defined in Eq. (2), where x_i represents the mole fraction of the component *i*.

$$SN = \frac{x_{H_2} - x_{CO_2}}{x_{CO} + x_{CO_2}} \quad (2)$$

The SN should be slightly above 2, that is considered the best condition for entering the methanol loop (Tountas et al., 2019). This condition also satisfies the commercial practice of maintaining a high CO:CO_2 ratio to increase the reaction rate and the CO_x conversion to methanol (Tountas et al., 2019).

2.1.4. Methanol production

Commercial methanol synthesis is carried out using a $CuO/ZnO/Al_2O_3$ catalyst at temperatures ranging from 200 to 300 °C and pressures from 50 to 100 bar (Ott et al., 2012). There are different reactor configurations, but fixed bed reactor with quench or multi-tubular reactor with cooling are mainly used for industrial applications (Leonzio, 2020). Commercially, the most used technology is the Lurgi externally-cooled, multi-tubular catalytic reactor (Leonzio, 2020). Methanol production reaction is limited by chemical equilibrium, and higher methanol yield is obtained at higher pressures and lower temperatures (Nimkar et al., 2017). Thus, the recycle of unreacted gas is necessary in order to achieve higher conversions (Lücking, 2017).

The methanol synthesis is modeled using the LHHW (Langmuir-Hinshelwood Hougen-Watson) kinetic model reformulated by Kiss et al. (2016) and implemented in Aspen Plus software. The model considers three main equilibrium reactions, namely R.1-R.3.

$$CO + 2H_2 \leftrightarrow CH_3OH \quad (\Delta H^o_{298K} = -90.77\, kJ/mol) \tag{R.1}$$

$$CO_2 + H_2 \leftrightarrow CO + H_2O \quad (\Delta H^o_{298K} = +41.21\, kJ/mol) \tag{R.2}$$

$$CO_2 + 3H_2 \leftrightarrow CH_3OH + H_2O \quad (\Delta H^o_{298K} = -49.16\, kJ/mol) \tag{R.3}$$

The corresponding rate equations for the kinetic model are Eqs. (3)–(5). The detailed description of the parameters can be found in (Kiss et al., 2016).

$$r_{CH_3OH} = k_A \frac{K_{CO}\left[f_{CO} f_{H_2}^{3/2} - f_{CH_3OH}/(K_A\sqrt{f_{H_2}})\right]}{(1 + K_{CO} f_{CO} + K_{CO_2} f_{CO_2})\left[\sqrt{f_{H_2}} + (K_{H_2O}/\sqrt{K_H}) f_{H_2O}\right]} \tag{3}$$

$$r_{CO} = r_{H_2O} = k_B \frac{K_{CO_2}[f_{CO_2} f_{H_2} - f_{H_2O} f_{CO}/K_B]}{(1 + K_{CO} f_{CO} + K_{CO_2} f_{CO_2})\left[\sqrt{f_{H_2}} + (K_{H_2O}/\sqrt{K_H}) f_{H_2O}\right]} \tag{4}$$

$$r_{CH_3OH} = r_{H_2O} = k_C \frac{K_{CO_2}\left[f_{CO} f_{H_2}^{3/2} - f_{H_2O} f_{CH_3OH}/(K_C f_{H_2}^{3/2})\right]}{(1 + K_{CO} f_{CO} + K_{CO_2} f_{CO_2})\left[\sqrt{f_{H_2}} + (K_{H_2O}/\sqrt{K_H}) f_{H_2O}\right]} \tag{5}$$

The validation of the kinetic model was performed using the data and conditions described in (Kiss et al., 2016). A sensitivity analysis at the reactor unit was performed, varying the temperature in the range of 200–300 °C and the pressure between 50 and 100 bar aiming to determine the best operatiing conditions.. As it has been reported in Kiss et al. (2016), the CO_2 and CO hydrogenation reactions are decreasing-volume reactions, thus, more CO is converted to methanol at higher pressures.

Yang et al. (2013) experimentally observed that, depending on the reaction temperature, the carbon atom incorporated into the MeOH molecule can originate from either CO or CO_2. As the temperature is lowered (below 240 °C), the preferred carbon source for the reaction changes from CO_2 to CO. In the same experiment, as the temperature is increased above 240 °C, the CO_2 carbon becomes the dominant reactant (Yang et al., 2013). Thus, as the syngas obtained from BLG contains a higher fraction of CO than CO_2, , the methanol formation is favored at lower temperatures. Based on a sensitivity analysis, a temperature of 210 °C and a pressure of 90 bar have been selected.

Finally, the purified syngas is compressed up to 90 bar and fed to the methanol synthesis loop. Therein, the syngas is heated up by the reactor effluent in a feed-effluent heat-exchanger (FEHE). Next, the syngas is fed to an isothermal reactor operating at 90 bar and 210 °C. The reactor outlet stream is a gaseous mixture of methanol, water and unconverted reactants. This mixture is cooled and flashed twice, first to 30 °C and 45 bar, and then to 3.5 bar, in order to separate the condensable products and the non-condensable reactants (Kiss et al., 2016). The condensed stream continues to a distillation column (DC) at atmospheric pressure, in which methanol is produced with purity over 99% wt. A fraction of the non-condensable stream is purged to avoid the built up of inerts, whereas the remainder is recycled back to the methanol synthesis loop.

2.1.5. Dimethyl ether production

The DME synthesis occurs via the indirect route, i.e. methanol dehydration (R.4) over an acid zeolite catalyst.

$$2CH_3OH \leftrightarrow CH_3OCH_3 + H_2O \quad (\Delta H^o_{298K} = -23.4\, kJ/mol) \tag{R.4}$$

The purified syngas initially passes through the same processes described for the methanol synthesis. Syngas is firstly compressed up to 90 bar and converted to methanol. Next, it continues to the DME synthesis loop, wherein an adiabatic DME reactor is used. The DME reaction is not highly exothermic and the temperature can be maintained by preheating the feed and running the reactor adiabatically (Al-Malah, 2017). For typical operating conditions of DME synthesis (250–400 °C), no significant side reactions are present. However, catalyst is significantly deactivated at temperatures above 400 °C and, for this reason, the reactor is designedto not exceeded this temperature. In this work, the feed temperature of methanol was defined using a design specification that limits the outlet temperature of the DME reactor, whereas maximizes the DME yield. The reaction rate equation (Eq. (6)) is given by (Bandiera and Naccache, 1991), which was adapted to Aspen Plus form by (Luyben, 2011; Al-Malah, 2017).

$$-r_{CH_3OH} = k_0 \exp(E/RT)\, p_{CH_3OH} \tag{6}$$

where k_0 = 0.336111 kmol/(m^3.s.Pa), E = 80.48 kJ/mol, and p_{CH3OH} is the partial pressure of methanol.

The reactor effluent is a mixture of water, DME and methanol, that is fed to a first distillation column (DC), which removes DME as distillate with a purity over 98.5% wt. Meanwhile, the bottoms stream is composed of water and methanol, which is fed to a second DC. The methanol that exits the second column as the distillate is recycled back to the reactor, while the water exits at the bottom.

2.1.6. Process resources and utility units

Different energy resources can be consumed in the integrated chemical plants shown in Fig. 1 in order to manufacture the marketed products (e.g. pulp, electricity, MeOH and DME). The utility systems involved are also shown in Fig. 1. The integration and sizing of one or more of utility systems and the selection of their operating conditions impact the energy consumption and the grade of the waste heat recovery.

2.2. Performance indicators

The conventional kraft pulp mill, based on the recovery boiler, is compared to other two proposed designs based on the upgraded BL gasification, aiming to produce either methanol or dimethyl ether. Moreover, two operation modes, namely mixed mode (i.e. importing both electricity and surplus biomass) or autonomous mode (i.e. only importing biomass), are considered for the integrated plants. The performance indicators adopted are described below.

2.2.1. Exergy analysis indicators

To assess the technical performance of the chemical processes, several indicators based on the First and Second Laws of Thermodynamics have been developed, including the concept of exergy. Exergy is related to the quality of energy. It is defined as the maximum available work that can be obtained from a thermodynamic system when it interacts with the environment by means of reversible processes until the equilibrium state is attained (mechanical, thermal and chemical) (Szargut et al., 1988). The exergy balance applied to a particular process unit or to a whole chemical process reveals how much of the maximum available energy provided by the feedstock is being utilized in a useful way by the systems (Kotas, 1995). The exergy method demonstrates the practical application of the Second Law of Thermodynamics in the analysis of industrial processes, e.g. revealing how and where exergy is used or destroyed..

The total exergy, B [kW] Eq. (7)), is the sum of four exergy components, i.e. the kinetic (K), potential (P), thermo-mechanical or physical (PH) and chemical (CH) components (Szargut et al., 1988), calculated by using Eqs. (7)–(11), respectively:

$$B = B^K + B^P + B^{PH} + B^{CH} \tag{7}$$

$$B^K = 0.5 \cdot m \cdot |\vec{v}|^2 \tag{8}$$

$$B^P = mgz \tag{9}$$

$$B^{PH} = m \cdot [h - h_0 - T_0(s - s_0)] \tag{10}$$

$$B^{CH} = n_{mix}\bar{b}^{CH} = n_{mix}\left[\sum_i x_i b_i^{CH} + R_u T_O \sum_i x_i \ln \gamma_i x_i\right] \tag{11}$$

where $\vec{v}$ and z are the velocity (vector, m/s) and elevation (scalar, m) of the system, respect to a stationary reference; m is the mass flow rate (kg/h); h and s are the specific enthalpy (kJ/kg) and entropy (kJ/kg-K), respectively; h_0 (kJ/kg) and s_0 (kJ/kg-K) are the specific enthalpy and entropy at standard conditions ($_{T_o = 298\ K\ and\ P_o = 1\ bar}$), respectively; x_i and γ_i are the mol fraction and the activity coefficient of the component i in the mixture; b_i^{CH} is the standard chemical exergy of component i (kJ/kg); R_u is the universal constant of gases, and n_{mix} is the total molar flow of the mixture (kmol/h).

Solid and liquid industrial fuels and other substances, such as biomass, are often mixtures of numerous chemical compounds, and it is considered that the ratio of the chemical exergy to the lower heating value ($\varphi = b^{CH}/LHV$) is invariant for pure chemical substances having the same mass ratios of chemicals constituents (H/C, O/C, N/C). Szargut et al. (1988) derived correlations that express the dependence of φ on those atomic ratios (Szargut et al., 1988). For biomass and other solid and dry organic substances composed of C, H, O and N, and 0.5 < O/C < 2, the value of φ can be calculated using Eq. (12), at standard conditions (P_o,T_o).

$$\varphi_{dry} = \frac{1.044 + 0.0160(H/C) - 0.3493(O/C)(1 + 0.0531(H/C)) + 0.0493(N/C)}{1 - 0.4124(O/C)} \tag{12}$$

Meanwhile, the lower heating value (LHV) is estimated based on the correlations reported by (Channiwala and Parikh, 2002), Eq. (13).

$$LHV = 349.1C + 1178.3H + 100.5S - 103.4O - 15.1N - 21.1ASH - 0.0894Hh_{lv} \tag{13}$$

where carbon (C), hydrogen (H), sulfur (S), oxygen (O), nitrogen (N) and ASH are the respective mass fractions of the components in the dry biomass, and h_{lv} is the enthalpy of evaporation of water at standard conditions (2,442.3 kJ/kg).

Table 2
Calculated lower heating value (LHV) and specific chemical exergy (b^{CH}) for wood, bark, black liquor, oil and pulp.

Material	LHV(MJ/kg dry)	b^{CH}(MJ/kg dry)
Wood	18.85	21.23
Bark	17.90	20.13
Black Liquor	10.28	12.08
Oil	40.56	43.38
Pulp	16.32	19.80

The calculated lower heating value (LHV) and specific chemical exergy (b^{CH}) of the materials considered in this work, are presented in Table 2.

The exergy balance formulation applied to a control volume operating in steady state allows to calculate the overall amount of irreversibility generated in the system, B_{dest}, according to Eq. (14):

$$B_{dest} = \sum_e m \cdot b - \sum_s m \cdot b + \int \left(1 - \frac{T_O}{T}\right)\delta Q - W_{useful} \tag{14}$$

where b is the specific exergy of the mass flows going through the control volume, Eq. (15):

$$b = (h - h_O) - T_O(s - s_O) + 1/2\left(m\upsilon^2\right) + mgz + b^{CH} \tag{15}$$

where h is the specific enthalpy, and s is the specific entropy. The terms related to the kinetic and potential exergy in Eq. (15) can be often neglected in comparison with the chemical and physical exergy of the mass flows. Q and W_{useful} are the rate of heat transfer and useful power, respectively, used to calculate the exergy related to the heat and work that enter or exit the control volume. The calculation of the physical and chemical exergies is performed via Excel add-ins embedded in Aspen Plus®.

In order to compare the different configurations, two exergy efficiency indicators are proposed, namely the rational exergy efficiency (Eq. (16)) and the relative exergy efficiency (Eq. (17)). The rational exergy efficiency defined in Eq. (16) assumes that the useful output of the integrated energy systems is the total exergy (B *in kW*), not only of the valuable products (MeOH or DME, and cellulose), but also of the by-products (CO_2, purge gas, power).

$$\eta_{\text{Rational}} = \frac{B_{useful,output}}{B_{input}} = 1 - \frac{B_{Dest}}{B_{input}} = 1 - \frac{B_{Dest}}{B_{oil} + B_{wood} + B_{chips} + W_{net}} \tag{16}$$

On the other hand, Eq. (17) quantifies the deviation from the minimum theoretical exergy consumption necessary to make up the main industrial products, i.e. pulp and chemicals.

$$\eta_{\text{Relative}} = \frac{B_{consumed,\ ideal}}{B_{consumed,\ actual}} = \frac{B_{chemical} + B_{pulp}}{B_{oil} + B_{wood} + B_{chips} + W_{net}} \tag{17}$$

In Eqs. (16) and (17), the overall exergy input corresponds to all the energy resources consumed, as depicted in Fig. 1. Moreover, the term B_{Dest} stands for the irreversibility rate, owed to the imperfection of the actual processes. This indicator can be estimated based on the unit and plant-wise level, considering all the subunits that compose the conventional pulp mill and the integrated chemical plants.

The renewability index, defined in Eq. (18), is used to evaluate the potential to offset (i) the utilization of non-renewable exergy resources B_{fossil}, (ii) the surplus deactivation exergy required for treating wastes $B_{deactivation}$(iii) the emissions of wastes that are not treated or deactivated $B_{emissions}$, and the exergy destroyed inside the system $B_{destroyed}$. This potential depends on the amount of renewable product $B_{product}$ derived from the conventional or the

integrated chemical plants (Velásquez et al., 2009).

$$\lambda = \frac{\sum B_{product}}{B_{fossil} + B_{destroyed} + B_{deactivation} + \sum B_{emissions}} \quad (18)$$

Accordingly, the renewability index indicates that the processes are:

- Environmentally unfavorable if $0 \leq \lambda < 1$.
- Internally and externally reversible with non-renewable inputs if $\lambda = 1$.
- Environmentally favorable if $\lambda > 1$.
- Thermodynamically reversible, consuming only renewable resources and producing no wastes when $\lambda \rightarrow \infty$.

The non-renewable exergy consumption (B_{fossil}) accounts for both the direct oil consumption and the indirect fossil exergy consumed at the upstream supply stages of oil (1.0298 kJ/kJ_{oil}) biomass (0.059 $kJ/kJ_{biomass}$) and electricity (0.3329 $kJ/kJ_{electricity}$), as reported in (Flórez-Orrego, 2014a, 2015). $B_{deactivation}$ considers the treatment of wastewater(1.7 kJ/kg_{H2O}) (Flórez-Orrego, 2014b).

2.2.2. CO_2 emissions assessment

Two CO_2 emissions balances were performed, namely, the overall and the net CO_2 emissions balances. The overall CO_2 emissions balance (Eq. (19)) is calculated by assuming that the emitted CO_2 (either fossil or biogenic) cannot be compensated by the avoided CO_2 that is captured in the syngas purification unit. In this case, biogenic emissions are accounted in the emissions balance of the plant, so that a more conservative case can be analyzed and means for mitigating these emissions can be also suggested. On the other hand, the net CO_2 emissions balance (Eq. (20)) assumes that the CO_2 molecules absorbed during the crops growth ($Biogenic^{direct}_{emissions}$) represent circular emissions, thus, this amount of CO_2 is subtracted from the overall CO_2 emissions. The indirect, non-renewable CO_2 emissions take into account those generated at the upstream supply chains of the electricity (62.09 g_{CO2}/kWh), wood (0.0043 g_{CO2}/kJ_{Wood}) and oil (0.0029 g_{CO2}/kJ_{Oil}) (Flórez-Orrego, 2014a 2015). The biogenic emissions comprise those derived from the combustion of woody biomass (bark and chips), which in turn depend on the biomass carbon content.

$$\begin{aligned} Overall\ CO_{2\,emissions} &= Fossil^{direct\ and\ indirect}_{emissions} + EE^{indirect}_{emissions} \\ &+ Wood^{direct\ and\ indirect}_{emissions} \\ &+ Oil^{direct\ and\ indirect}_{emissions} - CO_{2\,captured} \end{aligned} \quad (19)$$

$$\begin{aligned} Net\ CO_{2\,emissions} &= Fossil^{direct\ and\ indirect}_{emissions} + EE^{indirect}_{emissions} \\ &+ Wood^{indirect}_{emissions} + Oil^{direct\ and\ indirect}_{emissions} - CO_{2\,captured} \end{aligned} \quad (20)$$

Eqs. (21) and (22) illustrate how the indirect and direct emissions are calculated, respectively.

$$m^{\ indirect}_{emissions}[kg_{CO2}/s] = m_{Fuel}\ [kg_{Fuel}/s] \times b_{Fuel}[kJ_{Fuel}/kg_{Fuel}] \times \frac{r_{Fuel}[g_{CO2}/kJ_{Fuel}]}{1000} \quad (21)$$

$$m^{\ direct}_{emissions}[kg_{CO2}/s] = m_{Fuel}\ [kg_{Fuel}/s] \times C_{Fuel}[kg_C/kg_{Fuel}] \times \frac{M_{CO2}}{M_C}[kg_{CO2}/kg_C] \quad (22)$$

where r (g_{CO2}/kJ_{fuel}) is the indirect CO_2 emissions factor, M is the molecular mass (kg/kmol), and C_{Fuel} (%) is*percentage of carbon* for each fuel reported in Table 1. The consumption of the biomass and fuels, as well as *LHV* and b^{ch} ($\varphi = b^{CH}/LHV$) were presented in Table 2.

2.3. Optimization framework and optimization problem definition

Due to the possibility of importing electricity at the expense of a reduction of the biomass consumption, a trade-off between additional fuel and electricity importmay arise when the best alternatives for the utility system are targeted. Thus, the shift to alternative production routes may lead to completely different integration approaches between the chemical units and the utility systems.

For this reason, the OSMOSE Lua framework Yoo et al., 2015) was used to calculate the minimum energy requirements (MER) of the chemical plants. Subsequently, this framework handles the mixed integer non-linear programming problem that determines the best energy technologies and their working conditions that reduce the operating costs. The ASPEN Plus® software is used to simulate the chemical plants and transfer data to OSMOSE Lua to build the MILP problem described in Eqs. (22)–(26). This problem is solved using the open source GLPK solver (Yoo et al., 2015). The integer variables, y_w, related to the presence or not of a utility unit, ω, and its respective continuous load factor, f_w, can be determined by the minimization of the objective function given by Eq. (22):

$$\min_{\substack{f_\omega,\ y_\omega \\ R_r,\ W}} \begin{bmatrix} f_{Chips}(B\cdot c)_{Chips} + f_{Wood}(B\cdot c)_{Wood} + f_{Oil}(B\cdot c)_{Oil} + f_{Power}(W\cdot c)_{Power} \\ -f_{Pulp}(B\cdot c)_{Pulp} - f_{NH_3}(B\cdot c)_{NH_3} - f_{CO_2}(m\cdot c)_{CO_2} \end{bmatrix} \quad (22)$$

Subject to:

Heat balance at the temperature interval *r:*

$$\sum_{\omega=1}^{N_\omega} f_\omega q_{\omega,r} + \sum_{i=1}^{N} Q_{i,r} + R_{r+1} - R_r = 0 \quad \forall r = 1\ ..\ N \quad (23)$$

Produced and consumed power balance:

$$\sum_{\omega=1}^{N_\omega} f_\omega W_\omega + \sum_{chemical\ units} W_{net} + W_{imp} - W_{\exp} = 0 \quad (24)$$

Utility unit existence and load:

$$f_{\min,\omega} y_\omega \leq f_\omega \leq f_{\max,\omega} y_\omega \quad \forall \omega = 1\ ..\ N_\omega \quad (25)$$

Solution feasibility (MER):

$$R_1 = 0, \quad R_{N+1} = 0, \quad R_r \geq 0\ and\ W_{imp} \geq 0, \quad W_{\exp} \geq 0 \quad (26)$$

In the Eqs. (23)–(26), N is the amount of intervals of temperature depending on the supply and the temperatures target of the various streams; Q is the heat exchanged between the process streams ($Q_{i,r} > 0$ for hot stream, $Q_{i,r} < 0$ for cold stream) and R is the heat flow rate cascaded from higher ($r+1$) to lower (r) intervals of temperature (kW). Moreover, N_ω is the number of units available in the utility systems; B is the resources exergy flow rate (kW). In addition, q are the cooling or heating flow rates supplied by the utility systems (kW), and W is the produced power by the utility systems, chemical processes or exported to/imported from the grid (kW).

The price (c) of feedstock (€ per kWh or kg), electricity consumed or marketable products (e.g. pulp, MeOH or DME) and byproducts (e.g. surplus power and CO_2 produced) are defined in Table 3.

2.4. Economic evaluation

The capital expenditure (CAPEX) of the main process equipments is correlated to a reference equipment capacity using scaling factors, as described in (Turton et al., 2018). Eq. (27) is used to estimate CAPEX of the black liquor gasifier, the air separation unit, the water gas shift reactors, and the CO_2 capture unit, according

Table 3
Market costs and selling prices for feedstock and products.

	Market cost/selling price	Refs.
Wood	0.013 €/kWh	AEA (2008); Moraes (2011)
Chips	0.016 €/kWh	AEA (2008); Moraes (2011)
Oil	0.018 €/kWh	Moraes (2011)
Electricity	0.06 €/kWh	Moraes (2011)
Pulp	0.144 €/kWh	CelluloseOnline (2018)
Methanol	0.065 €/kWh	BSE Engineering (2017)
Dimethyl ether	0.07015 €/kWh	Fasihi and Breyer (2017)
CO_2	0.0084 €/kg	Florez-Orrego et al. (2017)

to the data reported in (Larson et al., 2000; Williams et al., 1995; Hamelinck and Faaij, 2002), and corrected for inflation using the suitable CEPCI index (Jenkins, 2020).

$$C_1 = C_0\left(\frac{S_1}{S_0}\right)^r \tag{27}$$

where S_0 represents the reference capacity with known capital cost C_0, and S_1 represents the actual capacity for which the capital cost C_1 is unknown. In addition, r is the power scaling factor, which may vary between 0.5 and 0.9 based on the type of process considered.

For cash flow calculations, plants lifespans of 20 years are assumed. Moreover, the total capital expenditure is divided between the first (60%) and second (40%) years. A decommissioning cost of 6% of the overall CAPEX is assumed. Meanwhile, the operating costs (OPEX) are estimated using the methodology proposed by Turton et al. (2018), except for the Kraft pulp mill (OPEX is assumed as 4% of the CAPEX Kangas et al., 2014). Due to the technological risk level, a contingency cost increment of 20% is considered (Turton et al., 2018).

2.4.1. Incremental financial analysis

Since further investments would be only justified by observable net benefits of the new arrangements over the existing configuration, an incremental approach is proposed to compare the economic attractiveness of the integrated chemical plants (Fig. 1) and the conventional process. This comparison is based on the incremental net present value (INPV), defined by Eq. (28) (Blank and Tarquin, 2011):

$$INPV = \sum_{n=1}^{N} \frac{\left[(\text{Rev} - \text{Exp})_{n,option\ B}\right] - \left[(\text{Rev} - \text{Exp})_{n,option\ A}\right]}{(1+i)^n} \tag{28}$$

where (Rev-Exp) is the net cash flow (i.e. revenues minus expenses), calculated at each (n) of the N yearly periods, during which both the alternative (B) and the reference (A) configurations must operate; i is the average interest rate.

Considering more stringent environmental regulations and expecting an increased risk perception associated to the proposed technologies (thus, higher interest rates for the investments), a sensitivity analysis of the variation of the INPV as a function of the carbon taxation (0–100 €/t_{CO2}) and the interest rate (0–21%) is performed.

Firstly, it is assumed that the prices of the feedstock and the products are set as constant. Next, the incremental financial analysis incorporates the uncertainty related to the acquisition and selling costs of the feedstock and fuels produced.

Thus, the second analysis is addressed using the Monte Carlo method, which allows simulating the stochastic variation of the commodities price profiles. This method is based on the generation of multiple trials to determine the value of a random variable, rendering it appropriate to estimate the uncertain variation of the INPV. A normal distribution with mean prices as those reported in Table 1 and a standard deviation of 30% is considered for the commodities. Also, 7,000 samples were considered in order to evaluate the proposed scenarios, which represents a reasonable amount of trials that consider outliers.

The INPV of the integrated chemical plants can be calculated along the lifespan of those facilities. Thereafter, the so-called '*likelihood of loss*', which measures the probability of achieving a negative incremental net present value, can be determined. The 'likelihood of loss' is the ratio between the negative INPV values among the sample trials divided by the total sample trials. Three hypothetical scenarios are considered:

- Deterministic carbon taxes (0–100 €/t_{CO2}) and interest rates (0–21%), along with stochastic prices of the commodities;
- Linearly-increasing carbon taxes over the lifespan (0–100 €/t_{CO2}), along with deterministic interest rates (0–21%) and stochastic prices of commodities;
- Stochastic carbon taxes and variable prices of the commodities, along with deterministic interest rates (0–21%).

The 'likelihood of loss' will depend on the fluctuations of the market prices of feedstock and products, the interest rates and carbon taxation (i.e. risk perception, technology readiness levels, governmental subventions, etc.), as well as on the effect of the depreciation of the money over time.

3. Results and discussion

In this section, the performance of the conventional kraft pulp mill is compared to that of the proposed chemical plants for methanol or dimethyl ether co-production.

3.1. Energy consumption remarks

The optimal process variables of the three chemical plants are compared in Table 4. According to these results, the integrated plants operating in autonomous mode present a consumption of fuel (wood chips) between five and seven times higher than in the cases in which the chemical plants operate in mixed mode. However, this lower chips consumption only comes at the expense of a considerable import of electricity when operating in the mixed mode. Accordingly, depending on the cost and the embodied emissions of the electricity generated in the electricity mix, there will be a trade-off between the self-generation of the utilities and the purchase of electricity from the grid. The MER of the conventional kraft pulp mill can be drastically cut down if a chemicalplant is integrated, which can be explained by the large extent of waste heat recovery and the capitalization on the residues, so that value added co-products can be produced.

According to Table 4, the integrated chemical plants operating under the mixed and autonomous modes present energy consumptions 14 and 38% higher than that of the conventional case (42.20 GJ/t_{Pulp}.) This fact can be explained by the increased biomass and electricity consumptions necessary to drive the compressors and other auxiliary equipment in the integrated chemical plants. More interestingly, an intensive import of Brazilian electricity during the mixed mode operation renders the MeOH and DME production less energy intensive if compared to the autonomous cases (see Table 4, cf. overall plant consumption). Furthermore, when the upstream supply chains of the feedstock consumed are accounted for, the extended exergy consumption increases up to 20%, compared to the plant-wide energy consumption. Those extended results are useful to issue public policies and decision-makers, since holistic comparisons of the impact of the supply chains in the bioproducts manufacturing sectors may be achieved, thus, ensuring a level playing field for comparisons between different MeOH or DME and pulp production routes.

Table 4
Optimal process variables of the studied production facilities.

Chemical plant	Conventional mill	Integrated chemical plants			
	Pulp	Pulp + MeOH		Pulp + DME	
Operation mode	BL combustion	Mixed	Auto	Mixed	Auto
Fuel input of the utility system	–	Electricity/Chips	Chips	Electricity/Chips	Chips
Feedstock wood consumption (GJ/t_{Pulp})	41.15	41.15	41.15	41.15	41.15
Consumption of chips for utility(GJ/t_{Pulp})	0	2.32	16.06	2.96	16.23
Consumption of electricity for utility (GJ/t_{Pulp})	0	3.3	0	3.18	0.00
Consumption of oil (GJ/t_{Pulp})	1.05	1.05	1.05	1.05	1.05
Overall plant consumption (GJ/t_{Pulp})	42.2	47.81	58.26	48.34	58.42
Extended plant consumption (GJ/t_{Pulp}) [1]	48.87	57.48	67.52	58.03	67.71
Power generation of Rankine cycle (GJ/t_{Pulp})[2]	4.26	1.74	5.22	1.86	5.22
Power demand of the chemical processes (GJ/t_{Pulp})					
- *KraftPulp Mill*	2.84	2.84	2.84	2.84	2.84
- *Black Liquor Treatment and Gasification*	1.20	1.20	1.20	1.20	1.20
- *Syngas Conditioning and MeOH/DME Synthesis*	0	0.30	0.30	0.31	0.31
Demand of ancillary power (GJ/t_{Pulp})[3]	0.14	0.06	0.24	0.07	0.24
Min. Cooling requirement (GJ/t_{Pulp})[4]	1.36	3.19	3.19	3.62	3.62
Min. Heating requirement (GJ/t_{Pulp})[4]	8.22	2.12	2.12	2.52	2.52
Consumption of biomass ($t_{Wood}/t_{Bioproduct+Pulp}$)	3.23	2.56	3.37	2.79	3.64
Syngas production (GJ/t_{Pulp})	0.00	9.36	9.36	9.36	9.36
MeOH or DME production yield (t/day)	0.00	289.62	289.62	208.19	208.19
Pulp production (t/day)	877.83	877.83	877.83	877.83	877.83
Marketable CO_2 production (kg/h)	0	32,714	32,714	32,714	32,714
Electricity export (kW)	878	0	0	0	0

1. It considers the extended exergy efficiency of the generation of electricity (55.68%), as well as of the oil (95.20%) and biomass (86.13%) production (Flórez-Orrego, 2014a; Flórez-Orrego et al., 2015); 2.Steam pressure levels 100, 12, 4 and 0.10 bar, steam superheating 200 °C; 3. Refrigeration and cooling tower; 4. MER of the chemical processes .

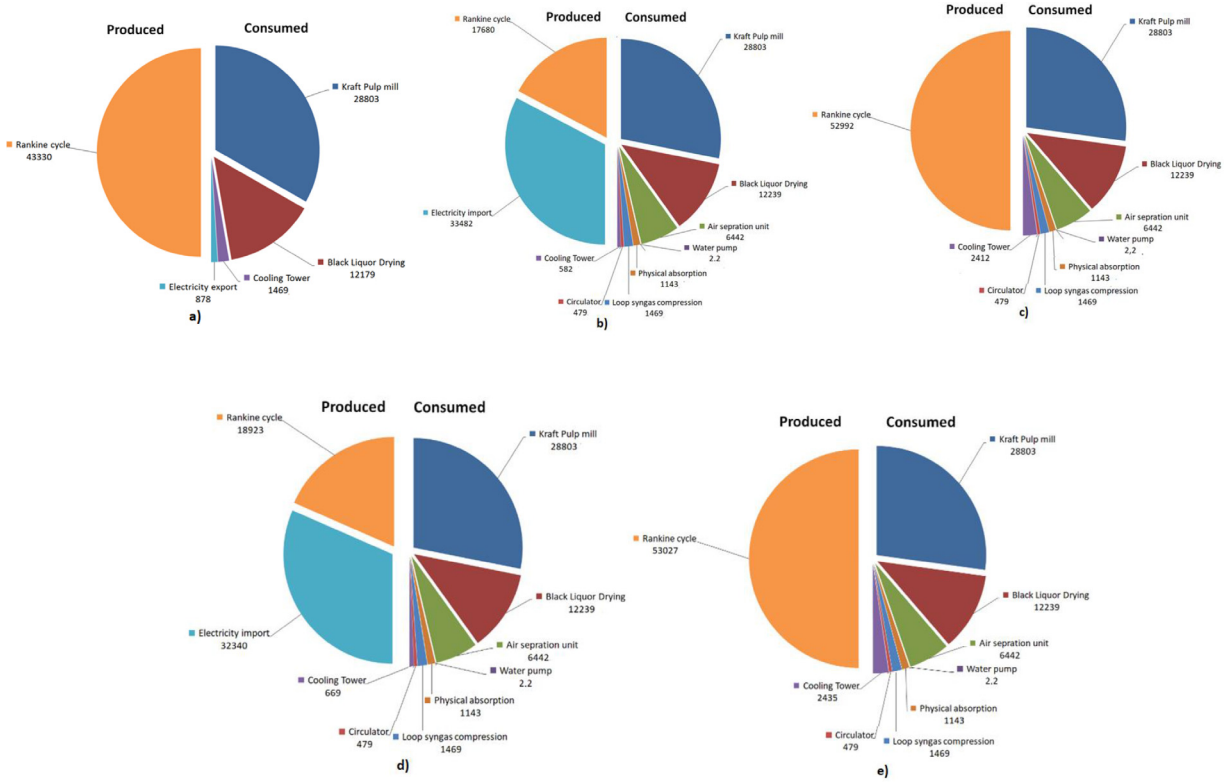


Fig. 2. Breakdown of the power consumption (kW): (a) Conventional case, (b) pulp&MeOH - Mixed operation mode, (c) pulp&MeOH - Autonomous operation mode, (d) pulp&DME - Mixed operation mode, (e) pulp&DME - Autonomous operation mode.

As for the power consumption breakdown (Fig. 2) in the integrated chemical plants, the kraft pulp mill accounts for more than half of the power used in those facilities. The electricity import in the integrated chemical plants that operate under the mixed mode can be as high as 63% of the total power demand in the plant. In addition, the autonomous cases (Fig. 2c and e) are responsible for the highest total power generation in the Rankine cycle, i.e. about 22% higher when compared to the conventional case. In the mixed cases (Fig. 2b and d), due to the diversification in the resources, the total power generation is reduced. This fact consequently affects other utility units, such as the cooling tower, which has a greater demand for au-

Table 5
Breakdown of the exergy destruction in the methanol and dimethyl ether integrated production routes (n.b. the utility systems are not included).

Units	MeOH	DME
Kraft pulp mill	73.52%	73.13%
Black liquor gasification unit	24.20%	24.07%
Syngas treatment unit	0.14%	0.14%
Gas purification unit	0.99%	0.98%
MeOH or DME conversion unit	1.15%	1.68%

tonomous operating modes due to the size of the Rankine cycle required.

On the other hand, the drying system based on the MVR process (Fig. 2) is responsible for 24% of the overall power consumed. This technology avoids the excessive steam consumption and the subsequent management of the condensate. Moreover, low-grade waste heat can be converted into power via the heat recovery network. Finally, the air separation unit consumes13% of the total power generated.

3.2. Energy integration analysis

Fig. 3a–e shows the energy integration integrated curves for all the chemical plants. Due to the introduction of additional energy conversion processes, a sharp modification ois observed for the curves of the integrated chemical plants (Fig. 3b–e). The integrated curves of the conventional kraft pulp mill are illustrated in (Fig. 3a). Due to the recovery of most of the waste heat available throughout the chemical plants, an otherwise rejected source of energy can be used to raise high pressure steam or preheat the process streams. Thus, there is an opportunity for reducing the energy import when integrating the conventional kraft process to the MeOH/DME chemical plants via the BL gasification process. Previous studies have already pointed out to this potential (Granacher et al., 2019), although no further analyses were presented. These results prove that the energy integration is useful for increasing the overall plant performance.

The substitution of the traditional multiple effect evaporation process by a MVR system also helps cutting down the quantity of steam consumed in the BL concentration.

Although the heating requirements of the conventional kraft pulp mill are fully supplied by the residual bark and black liquor combustion, it only comes at the expense of a higher quantity of exergy destruction per unit of product (pulp), if compared to the integrated plants operating under the mixed mode (Fig. 3b and d). The effect of surplus fuel (chips) consumption can be also observed in the curves corresponding to the integrated kraft pulp and MeOH or DME plants (Fig. 3c and e), as the span of the heat exchange (i.e. along the abscissa, H) becomes wider than in the case of electricity import (Fig. 3b and d).

It is worth noticing that these results are location-dependent, since more economically and environmentally friendly electricity mixes, such as that corresponding to the highly renewable Brazilian electricity mix, should be involved to take advantage of their higher generation efficiency.

3.3. Exergy-based indicators

Table 5 shows the breakdown of the exergy destruction for the two integrated chemical plants (i.e. with MeOH or DME co-production), excluding the utility systems. It is noteworthy that the distribution of the exergy destruction in the utility systems depends on the operation mode adopted (mixed or autonomous).

As it can be observed from Table 5, the kraft pulp mill holds the highest share of the exergy destruction, which can be explained by the energy intensive chemical processes necessary to separate pulp from wood and convert it into a commercial product. The black liquor gasification unit accounts for about 24% of the exergy destruction in the integrated plants, due to the reduction and oxidation reactions, while burning char at high temperatures to provide the energy demanded by those reactions (Domingos et al., 2021). In the syngas purification unit, the exergy destruction can be explained by the throttling for solvent regeneration, and the CO_2 separation and subsequent intercooled recompression. Finally, the exergy destruction in the methanol and dimethyl ether loops are mainly owed to the chemical reactions, recycled streams mixing and intercooling.

According to Table 6, the average exergy efficiency calculated for the conventional scenario is lower than those calculated for the integrated plants for MeOH and DME co-production. These results are strongly in agreement with those reported by Van Rens et al. (2011), who performed an exergy analysis for production of MeOH and DME using wood as feedstock via gasification process (Van Rens et al., 2011).

Among all the configurations studied, the integrated chemical plants operating under the mixed mode present the highest exergy efficiencies. These findings reinforce the idea that the electricity import, whether available, may help reducing the irreversibility in the manufacturing of bioproducts. Although intensive industrial utilization of the Brazilian electricity mix may bring about further challenges to electricity generation companies, the intensive electricity import in the integrated biorefineries has many potential opportunities that have not been fully exploited.

Meanwhile, a radical drop of up to 20% in the exergy efficiency is expected when incorporating the performance of the upstream supply chains, due to the inclusion of the irreversibility associated to the feedstock and fuels acquisition. Looking at the extended exergy destruction of the conventional kraft pulp mill and of the integrated chemical plants operating in mixed operation mode (around 31 GJ/t_{Pulp}), it becomes evident that the intensive consumption of biomass of the designs operating under autonomous mode becomes a decisive factor, as itinfluences both the irreversibilities (41 GJ/t_{Pulp}) and the carbon footprint associated to the co-production of chemicals. In other words, the apparent benefits of the biorefinery concept must be always weighed in the light of the extended production route, and not only from the plant-wide perspective.

3.4. Environmental analysis

Table 7 shows a detailed CO_2 emissions balance for all the configurations studied. In this figure, the avoided biogenic CO_2 emissions are associated to the carbon capture process in the syngas purification unit of the integrated chemical plants. The direct fossil CO_2 emissions are related to the oil consumption in the lime kiln, whereas the indirect fossil CO_2 emissions are associated to the biomass, electricity and oil supply chains. In fact, the largest share of indirect emissions belongs to the biomass acquisition and transportation steps, which is even higher than the direct emissions when burning by the fuel oil in the lime kiln. The present analysis reveals environmental issues that otherwise may remain unveiled if the biomass and electricity use were wrongly considered *emissions free*.

As a result, the overall emission balance achieves 1.97 t_{CO2}/t_{Pulp} in the conventional mill. Those values are much lower, even negative, in the case of the combined pulp and MeOH production plant (-0.07 t_{CO2}/t_{Pulp}), and the facilities coproducing pulp and DME (-0.02 t_{CO2}/t_{Pulp}) when running under the mixed operating mode. The negative values point towards a potential depletion of the CO_2 in the atmosphere by co-producing other chemicals in the integrated kraft pulp mills. The captured biogenic emissions offsets

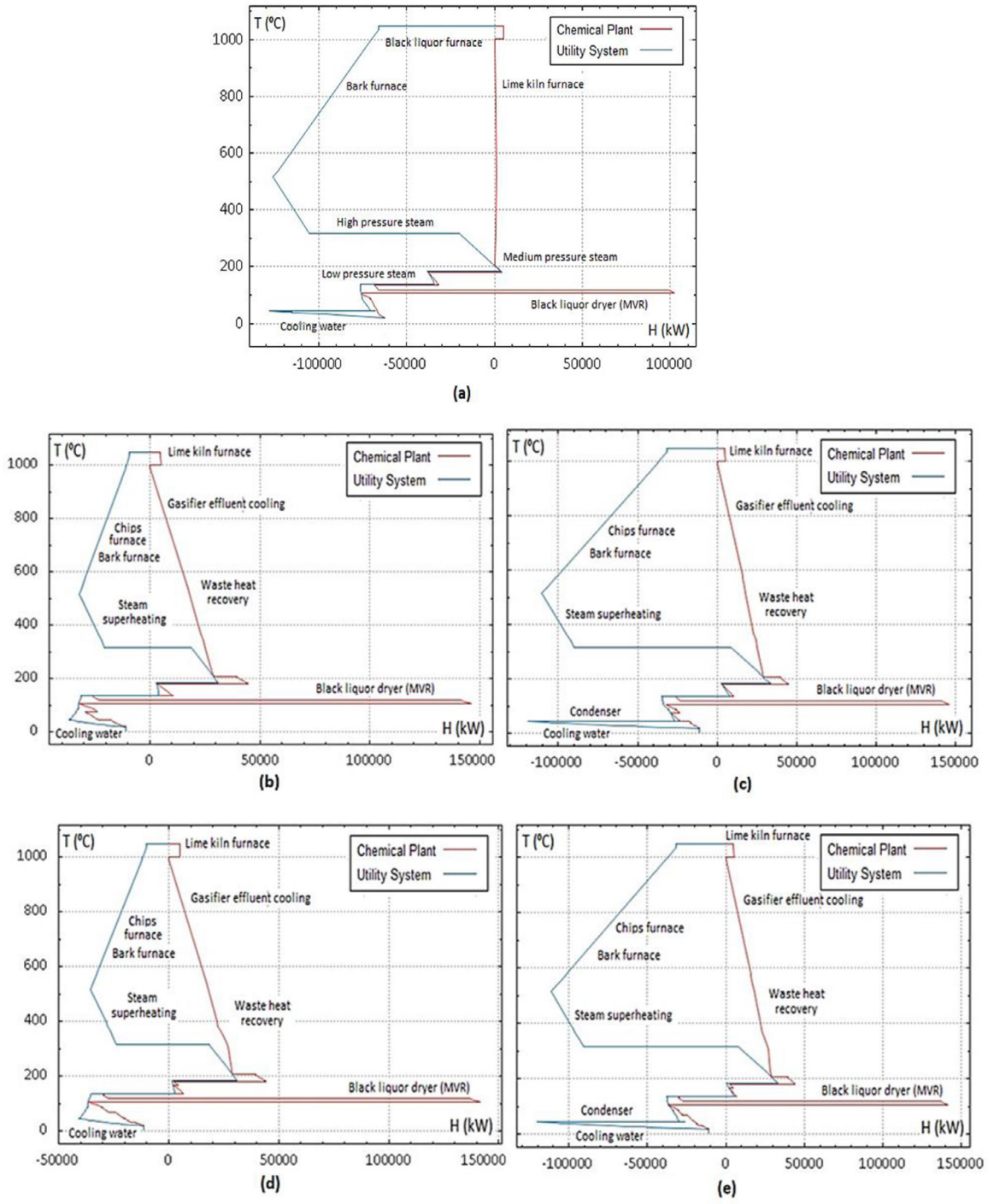


Fig. 3. Integrated composite curves: (a) Conventional kraft pulp mill; MeOH production under mixed (b) or autonomous (c) operating modes; and DME production under mixed (d) or autonomous (e) operating modes. Enthalpy flow rate (H in kW), Temperature (T in °C).

Table 6
Exergy performance indicators for the conventional and the conventional and the integrated chemicals production plants.

	Conventional mill	Integrated chemical plants			
Chemical plant	Pulp	Pulp + MeOH		Pulp + DME	
Operation mode	BL combustion	Mixed	Auto	Mixed	Auto
Rational exergy efficiency (%)	42.5	55.0	45.1	54.0	44.7
Extended rational exergy efficiency (%)	36.7	45.7	38.9	45.0	38.6
Relative exergy efficiency (%)	42.3	52.8	43.3	52.0	43.0
Extended relative exergy efficiency (%)	36.5	43.9	37.4	43.3	37.1
Exergy destruction (GJ/t_{Pulp})	24.3	21.5	32.0	22.2	32.3
Extended exergy destruction (GJ/t_{Pulp})	30.9	31.2	41.2	31.9	41.6

Table 7
Fossil, biogenic, direct and indirect CO_2 emissions of the conventional and the integrated chemicals production facilities.

	Conventional mill	Integrated chemical plants			
Chemical plant	Pulp	Pulp + MeOH		Pulp + DME	
Operation mode	BL combustion	Mixed	Auto	Mixed	Auto
Direct fossil CO_2 emitted (t_{CO2}/t_{Pulp})	0.08	0.08	0.08	0.08	0.08
Indirect fossil CO_2 emitted(t_{CO2}/t_{Pulp})	0.18	0.25	0.25	0.25	0.25
Total fossil CO_2 emitted (t_{CO2}/t_{Pulp})	0.26	0.33	0.33	0.33	0.33
Avoided biogenic CO_2 emissions (t_{CO2}/t_{Pulp})	0.00	−0.89	−0.89	−0.89	−0.89
Direct biogenic CO_2 emitted (t_{CO2}/t_{Pulp})	1.71	0.49	1.67	0.55	1.69
Overall* CO_2 *emissions balance (t_{CO2}/t_{Pulp})	**1.97**	**−0.07**	**1.11**	**−0.02**	**1.13**
Net* CO_2 *emissions balance (t_{CO2}/t_{Pulp})	**0.26**	**−0.57**	**−0.56**	**−0.57**	**−0.56**

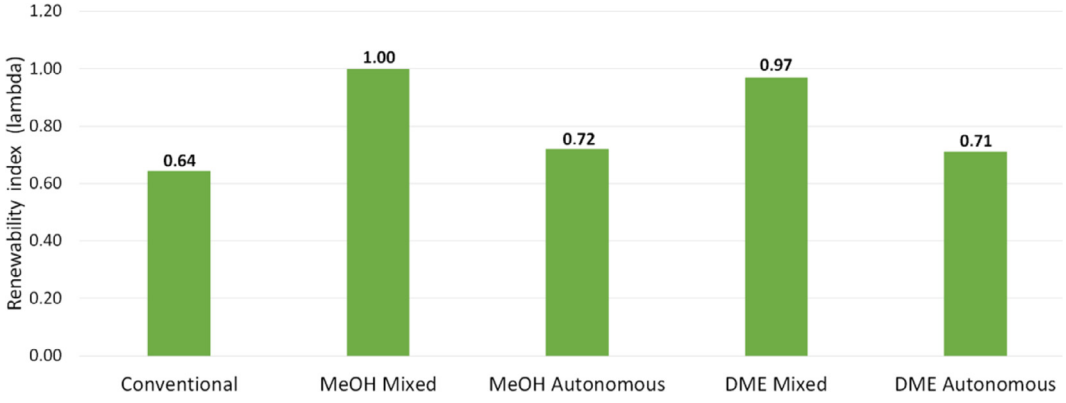


Fig. 4. Renewability exergy indicator for the conventional and the integrated chemical production plants.

the indirect CO_2 emissions from the feedstock supply chains with the aid of the intensive import of electricity.

Fig. 4 summarizes the results for the renewability exergy index. Compared to the conventional kraft pulp mill, in which this indicator achieves 0.64, the scenarios based on the integrated chemical plants running either on mixed (0.97) or autonomous (0.71) modes achieve higher renewability exergy indexes, with the co-production of pulp and methanol even attaining a renewability index of 1. These lambda values are similar to those reported in the literature for ethanol and sugar production from sugarcane in Brazil (0.66 to 0.72) (Velásquez et al., 2013). For this reason, the exergy embodied in the production of MeOH or DME by using BL gasification would not suffice to compensate for the exergy losses, the irreversibility and the fossil fuel consumption in the chemical plants studied. Similarly to the exergy efficiency behavior, the larger the electricity import, the higher the lambda index, which supports the use of electricity from the Brazilian electricity mix as a major driver of more sustainable biorefineries.

3.5. Economic analysis

According to Table 8, the integrated pulp and chemicals production plants running either on mixed and autonomous operating modes have 12–15% higher operating revenues owed to the other value-added products (i.e. apart from the pulp).

The breakdown of the capital expenditure shown in Fig. 5 implies that the operation under the autonomous mode requires bulkier equipment, such as larger chips furnaces and steam network components, compared to the plants running on the mixed operating mode. For this reason, the intensive use of electricity seems to be the most promising approach in terms of capital expenditure, provided that inexpensive and environmentally friendly energy resources are used in its generation.

3.5.1. Sensitivity analysis to carbon tax and interest rate

The attractiveness of the proposed setups will depend on certain financial parameters that affect the incremental net present value (INPV). For instance, even if the price of the commodities were considered as constant, any change in the carbon taxations and interest rates may alter the economic feasibility of a formerly viable solution. Accordingly, Fig. 6a–d show the variation of the INPV for all the integrated chemical plants as a function of the carbon tax (0–100 €/t_{CO2}) and the interest rate (0–21%).

As expected, more attractive financial conditions can be achieved for lower interest rates and higher carbon taxations. More specifically, for the integrated plants co-producing MeOH or DME and operating under the mixed mode, positive INPV values can be found for carbon taxes ranging between 20 and 70 €/t_{CO2}, depending on the interest rate adopted (see Fig. 6a and c). On the other hand, for the same integrated plants operating under the autonomous mode, the INPV cannot achieve positive values for DME production (Fig. 6d); whereas for the MeOH production, the INPV is positive only when the carbon tax becomes higher than 95 €/t_{CO2} and the interest rate is lower than 3%.

In this regard, middle-to-high carbon taxations (20–100 €/t_{CO2}) may render MeOH or DME co-production attractive in the Brazilian context of a highly renewable electricity mix. This feasibility range is in agreement with that suggested by the High-Level Commission on Carbon Prices, which estimated that a global CO_2 price of 50–100 USD/t_{CO2} would be needed by 2030 (WBG, 2020) to achieve the goals of the Paris Agreement.

3.5.2. Incremental financial analysis under uncertainty of the prices of the commodities

The aforementioned results assumed that the prices of the commodities could be considered as constant. However, in practice, those values are not deterministically defined, since they depend on the volatility of the market. In view of this circumstance, it is

Table 8
Annualized capital costs and net plant revenues for the the conventional and the integrated chemical production plants.

	Conventional mill	Integrated chemical plants			
Chemical plant	Pulp	Pulp + MeOH		Pulp + DME	
Operation mode	BL combustion	Mixed	Auto	Mixed	Auto
Operating Incomes (€/ t_{Pulp})	714.61	854.12	854.12	863.07	863.07
Operating Costs (€/t_{Pulp})	−153.49	−218.72	−224.89	−219.69	−225.62
Operating Revenues (€/t_{Pulp})[1]	561.12	635.40	629.24	643.37	637.45
Annualized capital cost (€/t_{Pulp})[2]	−281.13	−320.86	−345.55	−322.26	−327.16
Net Plant Revenues (€/t_{Pulp})[3]	280.00	**314.54**	**283.69**	**321.12**	**310.29**

1. The operating revenues are estimated as the difference between the gross operating incomes minus the operating cost; 2. The investment costs of pulp mill are scaled (0.6 factor) according to Börjesson (2015), other CAPEX costs calculated by means of the methodology of (Turton et al., 2018); 3. The net revenues are calculated as the operating revenues subtracting the yearly investment cost.

Table 9
Probability that the INPV is negative (*likelihood of loss* in%) as a function of the interest rate, for deterministic carbon taxes and stochastic prices of the commodities (scenario (a)) for an integrated pulp and MeOH chemical plant operating under the mixed mode.

CO_2 tax (€/t_{CO2}) →	0	10	20	30	40	50	60	70	80	90	100
0%	97.70	85.81	59.07	25.93	6.44	0.76	0.07	0.00	0.00	0.00	0.00
3%	99.59	95.26	78.46	46.97	17.63	3.83	0.47	0.00	0.00	0.00	0.00
6%	99.90	98.77	91.19	70.77	38.40	13.86	3.19	0.40	0.01	0.00	0.00
9%	99.99	99.57	97.34	87.64	64.73	36.11	12.40	2.59	0.33	0.01	0.00
12%	100.00	99.90	99.47	95.70	84.00	59.30	31.13	11.66	2.81	0.46	0.01
15%	100.00	100.00	99.83	98.51	93.44	79.50	54.66	28.61	11.37	2.61	0.37
18%	100.00	100.00	99.91	99.53	97.44	89.57	74.84	50.96	25.69	9.76	2.77
21%	100.00	100.00	99.97	99.87	99.06	95.53	86.96	69.01	45.61	23.50	9.31

i (%)

Table 10
Probability that INPV is negative (*likelihood of loss* in%) as a function of the interest rate, for deterministic carbon taxes and stochastic prices of commodities (scenario (a)) for an integrated pulp and MeOH chemical plant operating under the autonomous mode.

CO_2 tax (€/t_{CO2}) →	0	10	20	30	40	50	60	70	80	90	100
0%	100.00	100.00	100.00	100.00	100.00	99.94	99.77	97.40	85.81	59.10	27.77
3%	100.00	100.00	100.00	100.00	100.00	100.00	99.97	99.83	98.10	89.86	67.20
6%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.89	99.14	93.99
9%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	99.93	99.56
12%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
15%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
18%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
21%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00

i (%)

Table 11
Probability that the INPV is negative (*likelihood of loss* in%) as a function of the interest rate, for deterministic carbon taxes and stochastic prices of commodities (scenario (a)) for an integrated pulp and DME chemical plant operating under the mixed mode.

CO_2 tax (€/t_{CO2}) →	0	10	20	30	40	50	60	70	80	90	100
0%	96.37	83.77	55.69	25.79	6.61	0.81	0.09	0.01	0.00	0.00	0.00
3%	98.87	94.10	76.47	47.96	18.24	4.39	0.53	0.03	0.00	0.00	0.00
6%	99.74	98.19	90.60	70.89	41.34	15.74	3.66	0.47	0.03	0.00	0.00
9%	99.97	99.57	97.03	88.16	65.70	37.47	14.79	3.80	0.63	0.04	0.00
12%	100.00	99.80	99.26	95.20	84.31	61.04	35.03	15.44	3.23	0.67	0.09
15%	100.00	99.96	99.80	98.40	93.61	80.60	59.00	33.30	13.44	3.93	0.86
18%	100.00	100.00	99.89	99.47	97.61	90.87	76.11	55.40	31.86	14.57	4.70
21%	100.00	99.99	99.96	99.90	98.97	96.24	88.41	72.19	52.03	28.90	13.41

i (%)

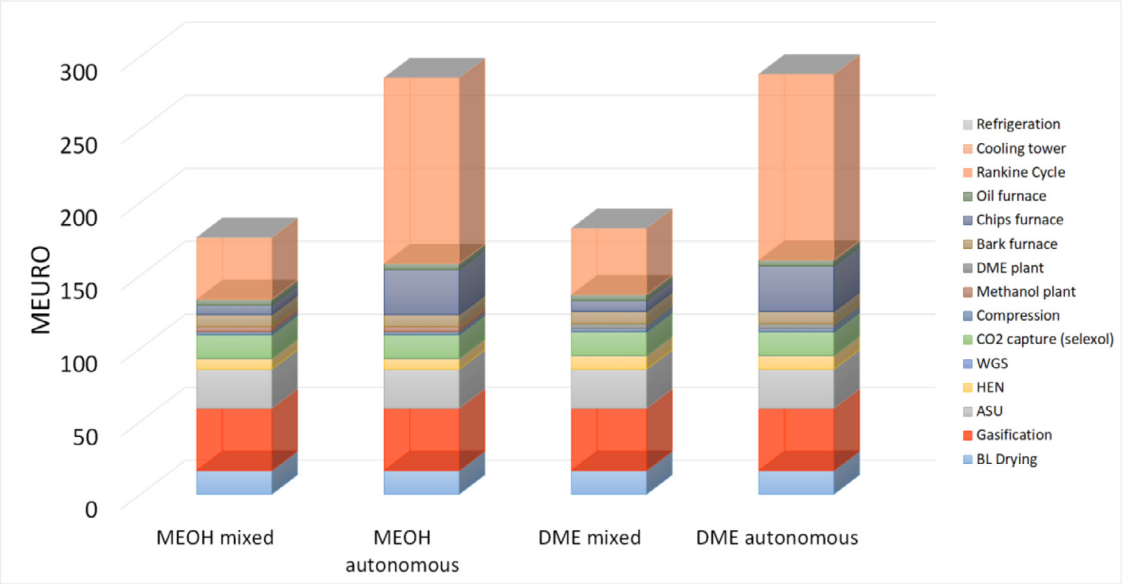


Fig. 5. Capital investment costs breakdown for the integrated chemical production plants. 20% of contingency is considered.

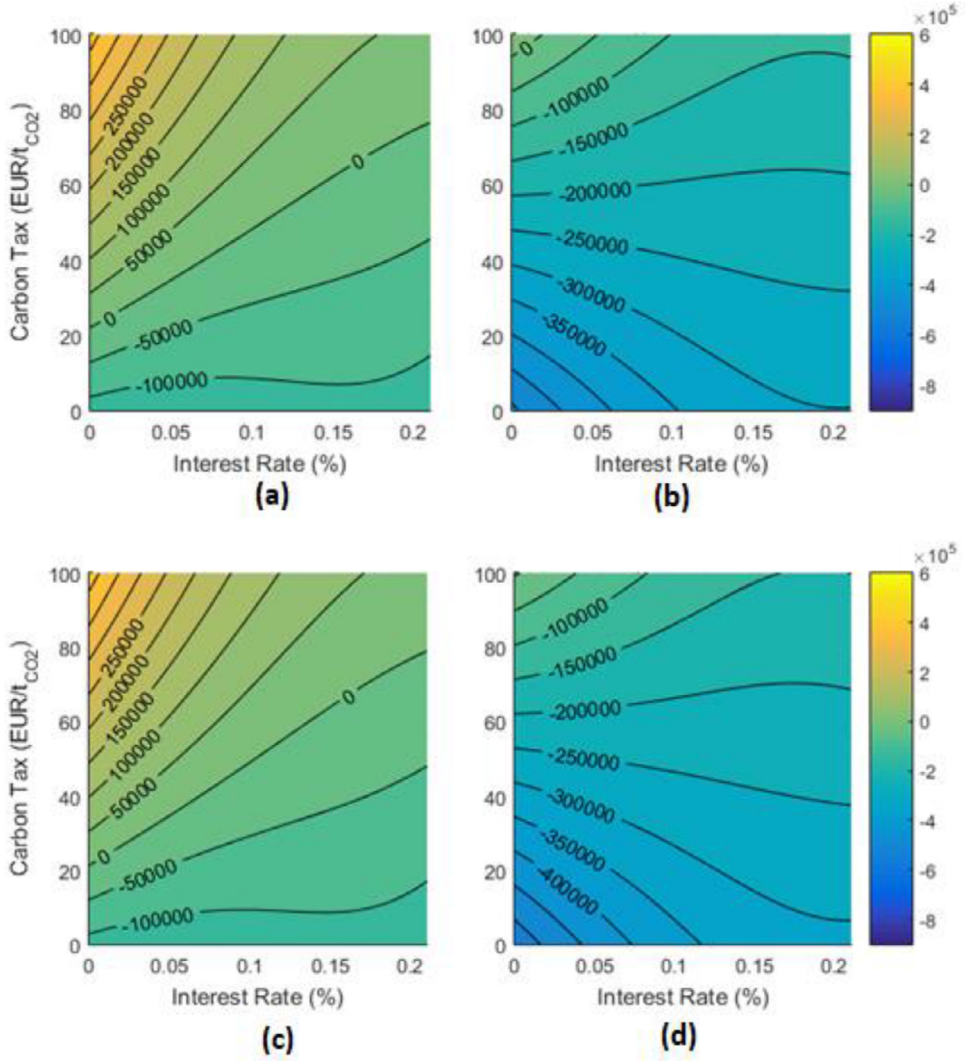


Fig. 6. Contour plots of INPV (€) variation: (a) integrated pulp and MeOH production operating in mixed mode, (b) integrated pulp and MeOH production operating in autonomous, (c) integrated pulp and DME production operating in mixed mode (d) integrated pulp and DME production operating in autonomous mode.

Table 12
Probability that INPV is negative (*likelihood of loss* in%) as a function of the interest rate, for deterministic carbon taxes and stochastic prices of commodities (scenario a) for an integrated pulp and DME chemical plant operating under the autonomous mode.

CO_2 tax (€/t_{CO2}) →	0	10	20	30	40	50	60	70	80	90	100
0%	100.00	100.00	100.00	100.00	100.00	100.00	99.94	98.30	92.16	73.71	43.64
3%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.84	99.09	94.69	80.36
6%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.93	99.59	97.20
9%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.97	99.81
12%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
15%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
18%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
21%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00

i (%)

Table 13
Probability that the INPV is negative (*likelihood of loss* in%) as a function of the interest rate considering linearly increasing carbon taxes and stochastic prices of the commodities (scenario (b)).

	pulp&MeOH - mixed mode	pulp&MeOH - autonomous mode	pulp&DME - mixed mode	pulp&DME - autonomous mode
0%	1.00	100.00	1.24	100.00
1%	2.37	100.00	2.16	99.99
2%	4.79	100.00	5.17	100.00
3%	9.37	100.00	10.46	100.00
4%	17.46	100.00	18.56	100.00
5%	27.77	100.00	29.49	100.00
6%	41.91	100.00	42.26	100.00
7%	56.33	100.00	55.40	100.00
8%	67.77	100.00	68.56	100.00
9%	77.80	100.00	78.61	100.00
10%	85.34	100.00	85.87	100.00
11%	91.99	100.00	91.20	100.00
12%	95.07	100.00	94.63	100.00
13%	96.93	100.00	96.69	100.00
14%	98.34	100.00	98.33	100.00
15%	99.11	100.00	98.89	100.00
16%	99.47	100.00	99.47	100.00
17%	99.69	100.00	99.66	100.00
18%	99.77	100.00	99.79	100.00
19%	99.89	100.00	99.81	100.00
20%	99.96	100.00	99.91	100.00
21%	99.97	100.00	99.97	100.00

i (%)

desirable to assess the effect of the stochastic variation of the price of those commodities on the financial attractiveness of the proposed plants. This analysis relies on the Monte Carlo simulations developed for the scenarios (a)–(c) described in Section 2.4.1.

The heat maps given in Tables 9–12 summarize the results obtained for the integrated chemical plant scenarios in which the prices of the commodities vary in a stochastic way, for a series of carbon taxes (0–100 €/t_{CO2}) and interest rates (0–21%). According to Table 9, for the integrated plant co-producing pulp and MeOH and operating under the mixed mode, there exists a large probability of the INPV being positive, depending on the interest rate adopted and for a region of middle to stringent carbon taxations (70–100 €/t_{CO2}). For the integrated plant co-producing pulp and DME and operating under the mixed mode, Table 11 shows that a economically feasible carbon tax starts at a higher carbon tax level (80–100 €/t_{CO2}). The set of Fischer interceptions (i.e. the transition region from green to yellow in the heat map of Tables 9 and 11) indicates the conditions at which the shift from a null *likelihood of loss* (i.e. financially feasible region) to a positive one (i.e. likely unfeasible region) occurs. In this way, the heat maps allow bounding the financially interesting conditions of the integrated biorefineries.

In this regard, at very low carbon taxes (< 10 €/t_{CO2}) typically found in the regional economies (Monge, 2018), even a very low interest rate would not be enough to shift the INPV to positive values. Meanwhile, at moderate carbon taxations, such as those imposed by Norway and Finland (~40–80 €/t_{CO2}) (Monge, 2018), the integrated chemical plant starts to bring about better INPV values if MeOH or DME are produced under the mixed operation mode. For the sake of comparison, Tables 10 and 12 shows that the inte-

Table 14
Probability that the INPV is negative (*likelihood of loss* in%) as a function of the interest rate, considering stochastic carbon taxes and prices of the commodities (scenario (c)).

i (%)	pulp&MeOH - mixed mode	pulp&MeOH - autonomous mode	pulp&DME - mixed mode	pulp&DME - autonomous mode
0%	1.49	99.96	1.67	99.96
1%	2.14	99.99	2.36	100.00
2%	3.24	100.00	3.64	99.99
3%	5.24	100.00	6.21	99.99
4%	8.37	100.00	8.89	100.00
5%	11.63	100.00	13.00	100.00
6%	16.81	100.00	18.30	100.00
7%	21.74	100.00	23.29	100.00
8%	28.81	100.00	31.86	100.00
9%	36.17	100.00	39.30	100.00
10%	44.74	100.00	47.31	100.00
11%	52.09	100.00	55.40	100.00
12%	60.14	100.00	61.77	100.00
13%	66.36	100.00	67.71	100.00
14%	72.81	100.00	75.71	100.00
15%	77.93	100.00	79.84	100.00
16%	82.31	100.00	84.59	100.00
17%	86.00	100.00	87.14	100.00
18%	88.89	100.00	89.96	100.00
19%	91.44	100.00	92.49	100.00
20%	92.83	100.00	93.80	100.00
21%	94.61	100.00	94.99	100.00

grated chemical plants co-producing MeOH or DME and operating under the autonomous modepresent relatively unfavorable INPVs, regardless of the value of the interest rates.

Meanwhile, the heat maps shown in Tables 13 and 14 summarize the results obtained for scenarios (b) and (c), described in Section 2.4.1. The former scenario (b) aims to represent the situation in which the carbon taxes linearly increase from 0 to 100 €/t_{CO2}, to allow for a smooth transition between the current and the future carbon taxed scenario. Similarly to the previous analyses, the prices of commodities are stochastically varied over the project lifespan. In the second scenario (c), all the previous assumptions remain the same, except for the carbon taxation, which turn out be also stochastic (i.e. randomly varied between 0 and 100 €/t_{CO2}). In both scenarios, the analysis is performed for an interest rate varying from 0% to 21%.

According to Tables 13 and 14, only the integrated chemical plants producing MeOH or DME operating under the mixed mode and at very low interest rates (~0%) presented close to favorable results in terms of likelihood of loss (i.e. the probability that INPV is negative). As a final remark, only the confluence of several factors, such as (*i*) equipment maturation and deployment; (*ii*) advantageous economies of scale; (*iii*) more stringent regulatory commitments pointing towards the decarbonization of chemicals and fuels; as well as (*iv*) attractive fiscal incentives and international cooperation may boost the integration of chemicals and fuels production via BLG in pulp and paper industries. These factors become more relevant if uncertainties are considered for the carbon taxations and the price of the commodities. The optimistic scenario is not at all unrealistic, considering that by 2040, more than two thousand large-scale CCS facilities are necessary to reach sustainable development goals and to comply with the Paris agreement, as stated by the International Energy Agency (IEA, 2019).

4. Conclusions

The thermo-environomic performances of two chemical production plants (MeOH or DME) integrated to a kraft pulp mill via the black liquor gasification process are compared to the performance of a conventional kraft pulp mill. By using the combined exergy and energy integration analyses, it was possible to work out the best alternatives of the utility systems. As a result, the exergy efficiencies of the conventional and integrated cases average 40% and 45%, respectively, whereas the overall emission balance varies from 1.97 to −0.07 t_{CO2}/t_{Pulp}, respectively. These negative emissions highlight the benefits of using alternative energy sources, such as biomass and electricity, in integrated chemical plants. The renewability exergy index turns out to be higher for the integrated chemical plants operating under the mixed mode, due to the reduction of the fossil energy consumption and the intensive import of the Brazilian electricity. The results pointed that the environmental and economic performances are highly related to the indirect emissions due to the acquisition of the energy resources and are subject to the adopted market prices for the commodities. The introduction of more stringent carbon taxes and more affordable biomass conversion technologies may encourage the deployment and diminish the financial risks associated to breakthrough technologies. Anyhow, the integrated black liquor gasification proves to be a feasible solution to co-produce chemicals, thus, revealing new business opportunities to decarbonize the bulk chemicals production.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Meire Ellen Gorete Ribeiro Domingos: Conceptualization, Methodology, Formal analysis, Software, Writing – original draft. **Daniel Flórez-Orrego:** Conceptualization, Methodology, Formal analysis, Writing – review & editing. **Moisés Teles dos Santos:** Project administration, Writing – review & editing, Supervision. **Silvio de Oliveira:** Formal analysis, Writing – review & editing. **François Maréchal:** Formal analysis, Writing – review & editing.

Acknowledgments

The first author thanks the Brazilian National Research Council for Scientific and Technological Development, CNPq (grant 142148/2019–9), and the Swiss Government Excellence Scholarship (ref. 2021.0235). Second author would like to thank the Colombian Administrative Department of Science, Technology and Innovation – Minciencias (COLCIENCIAS Grant 1.128.416.066–646/2014). The fourth author thanks CNPq (grant 306484/2020–0).

References

Adams Ii, T.A., Salkuyeh, Y.K., Nease, J., 2014. Processes and simulations for solvent-based CO_2 capture and syngas cleanup. Reactor and Process Design in Sustainable Energy Technology. Elsevier.

AEA. 2008. Excel Calculation_for_Wood_Fuel_Parameters_16_EN KLIMAACTIV [Online]. Vienna: austrian Energy Agency. [Accessed 19 january 2019] Available at: https://www.klimaaktiv.at/erneuerbare/energieholz/werkzeuge-und-hilfsmittel/kenndatenkalkulation.html.

Al-Malah, K.I., 2017. Aspen Plus: Chemical Engineering Applications. Wiley Online Library.

Bachmann, D.L., 2009. Berchmarking of pulp mill processes. O Papel 8, 66–70.

Bandiera, J., Naccache, C., 1991. Kinetics of methanol dehydration on dealuminated H-mordenite: model with acid and basic active centres. Appl. Catal. 69, 139–148.

Blank, L., Tarquin, A., 2011. Engineering Economy. Mc Graw Hill, El Paso, Texas.

Boloy, R.A.M., Silveira, J.L., Tuna, C.E., Coronado, C.R., Antunes, J.S, 2011. Ecological impacts from syngas burning in internal combustion engine: technical and economic aspects. Renew. Sustain. Energy Rev. 15, 5194–5201.

Börjesson, M., Ahlgren, E. 2015. Pulp and paper industry – energy technology system analysis program – ETSAP, IEA ETSAP - Technology Brief I07-May 2015. IEA. Available at: https://iea-etsap.org/E-TechDS/HIGHLIGHTS%20PDF/I07_Pulp&Paper%20May2015_%201st%20page.pdf.

BSE Engineering, 2017. Small scale methanol plants a chance for re-industrialisation. In: Proceedings of the International Methanol Conference. Available.

Carvalho, L., Lundgren, J., Wetterlund, E., Wolf, J., Furusjö, E., 2018. Methanol production via black liquor co-gasification with expanded raw material base – techno-economic assessment. Appl. Energy 225, 570–584.

Channiwala, S.A., Parikh, P.P., 2002. A unified correlation for estimating HHV of solid, liquid and gaseous fuels. Fuel 81, 1051–1063.

CeluloseOnline. 2018. Preço da celulose cresce em ritmo mais lento [Online]. Available: https://www.celuloseonline.com.br/preco-da-celulose-cresce-em-ritmo-mais-lento/ [Accessed December 20 2018].

Darmawan, A., Hardi, F., Yoshikawa, K., Aziz, M., Tokimatsu, K., 2017a. Enhanced process integration of black liquor evaporation , gasification , and combined cycle. Appl. Energy 204, 1035–1042.

Darmawan, A., Hardi, F., Yoshikawa, K., Aziz, M., Tokimatsu, K., 2017b. Enhanced process integration of entrained flow gasification and combined cycle: modeling and simulation using aspen plus. Energy Proced. 105, 303–308.

Dellavedova, M., Derudi, M., Biesuz, R., Lunghi, A., Rota, R., 2012. On the gasification of biomass: data analysis and regressions. Process Saf. Environ. Prot. 90, 246–254.

Demirbas, A., 2006. Theoretical heating values and impacts of pure compounds and fuels. Energy Sour. Part A Recovery Util. Environ. Eff. 28, 459–467.

Domingos,M.E.G.R., Florez-Orrego, D.A., Teles dos Santos, M., Oliveira Junior, S. 2019. Process integration enhancement via black liquor gasification in the Kraft pulp industry. Process system engineering - PSE-BR2019, Rio de Janeiro, Brazil.

Domingos, M.E.G., Flórez-Orrego, D., Teles dos Santos, M., Oliveira Junior, S., 2021. Comparative assessment of black liquor upgraded gasification in integrated kraft pulp, methanol and dimethyl ether production plants. Comput.-Aided Chem. En. 50, 25–30. doi:10.1016/B978-0-323-88506-5.50005-X.

Domingos, M.E.G.R., Florez-Orrego, D., Teles Dos Santos, M., Velasquez, H., De Oliveira Junior, S., 2021. Exergy and environmental analysis of black liquor upgrading gasification in an integrated kraft pulp and ammonia production plant. Int. J. Exergy 35 (1). doi:10.1504/IJEX.2021.115083.

Ekbom, T., Lindblom, M., Berglin, N. & Ahlvik, P. 2003. Technical and commercial feasibility study of black liquor gasification with methanol/DME production as motor fuels for automotive uses-BLGMF. Nykomb Synergetics AB, Chemrec, Volvo, Ecotraffic, OKQ8, STFi. Available at: https://cordis.europa.eu/project/id/4.1030-Z-01-087.

Ericsson, K. & Nilsson, L.J.. 2018. Climate innovations in the paper industry: prospects for decarbonisation. Available at: https://static1.squarespace.com/static/59f0cb986957da5faf64971e/t/5dc1acfb29bc520c15c858fc/1572973823092/%28updated%29D2.4+Climate+innovations+in+the+paper+industry.pdf.

Fasihi, M., Breyer, C., 2017. Synthetic methanol and dimethyl ether production based on hybrid PV-wind power plants. In: Proceedings of the 11th International Renewable Energy Storage Conference. Düsseldorf.

Fernández-Dacosta, C., Shen, L., Schakel, W., Ramirez, A., Kramer, G.J., 2019. Potential and challenges of low-carbon energy options: comparative assessment of alternative fuels for the transport sector. Appl. Energy 236, 590–606.

Ferreira, E.T.D.F., Balestieri, J.A.P., 2015. Black liquor gasification combined cycle with CO_2 capture – technical and economic analysis. Appl. Therm. Eng. 75, 371–383.

Florez-Orrego, D., Sharma, S., Oliveira, S., Marechal, F., 2017. Combined exergy analysis and energy integration for design optimization of nitrogen fertilizer plants. In: Proceedings of the 30th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems. San Diego, United States of America ECOS 2017, July 2 - 6San Diego State University, 200.

Flórez-Orrego, D., Silva, J.A.M., Velásquez, H., Oliveira, S., 2015. Renewable and non-renewable exergy costs and CO2 emissions in the production of fuels for Brazilian transportation sector. Energy 88, 18–36.

Flórez-Orrego, D., Silva, J.A.M., Oliveira, S., 2014a. Renewable and non-renewable exergy cost and specific CO_2 emission of electricity generation: the Brazilian case. Energy Convers. Manag. 85, 619–629.

Flórez-Orrego, D.A, 2014b. Thermodynamic and environmental comparison (CO_2) emissions of the route of the production and end use of vehicle fuels derived from petroleum, natural gas, biofuels, hydrogen and electricity MSc Thesis Dissertation.

Foelkel, C., 2017. The Eucalyptus Wood For Pulp Production – Understanding the Construction of the Indicator of Specific Consumption of Wood for Kraft Pulp Production. Eucalyptus Online Book & Newsletter [In Portuguese].

Fornell, R., Berntsson, T., Åsblad, A., 2013. Techno-economic analysis of a kraft pulp-mill-based biorefinery producing both ethanol and dimethyl ether. Energy 50, 83–92.

Gavrilescu, D., 2008. Energy from biomass in pulp and paper mills. Environ. Eng. Manag. J. 7, 537–546.

Global Combustion Systems. 2019. Oil fuel properties [Online]. Available: http://www.globalcombustion.com/oil-fuel-properties/ [Accessed 20 December 2018].

Granacher, J., Maréchal, F., Celebi, A.D., Kermani, M., 2019. Potential of Hydrothermal Black Liquor Gasification Integrated in Pulp Production Plant. Silesian University of Technology.

Häggström, C., Öhrman, O., Rownaghi, A.A., Hedlund, J., Gebart, R., 2012. Catalytic methanol synthesis via black liquor gasification. Fuel Proces. Technol. 94, 10–15.

Hamelinck, C.N., Faaij, A.P., 2002. Future prospects for production of methanol and hydrogen from biomass. J. Power Sour. 111, 1–22.

Harris, A.T., Riddlestone, S., Bell, Z., Hartwell, P.R., 2008. Towards zero emission pulp and paper production : the bioregional minimill. J. Clean. Prod. 16, 1971–1979.

Huet, M., 2015. Valorisation hydrothermales de la liqueur noire à des fins énergétiques et de chimie verte PhD Thesis. Université Grenoble Alpes.

IEA, 2007. Black liquor gasification - summary and conclusions. In: Proceedings of the IEA bioenergy ExCo54 Workshop. International Energy Agency Available at: https://www.ieabioenergy.com/blog/publications/exco54-workshop-black-liquor-gasification-summary-and-conclusions/.

IEA, 2019. World Energy Outlook 2019. IEA, Paris Available at: https://www.iea.org/reports/world-energy-outlook-2019.

IEA, 2020. Tracking Industry 2020. IEA, Paris Available at: https://www.iea.org/reports/tracking-industry-2020.

Jenkins, S. 2020. 2019 chemical engineering plant cost index annual average [Online]. Available: https://www.chemengonline.com/2019-chemical-engineering-plant-cost-index-annual-average/ [Accessed 08/02/2021 2021].

Kangas, P., Kaijaluoto, S., Määttänen, M., 2014. Evaluation of future pulp mill concepts – reference model of a modern Nordic kraft pulp mill. Nord. Pulp Paper Res. J. 29, 620–634.

Kiss, A.A., Pragt, J., Vos, H., Bargeman, G., De Groot, M., 2016. Novel efficient process for methanol synthesis by CO_2 hydrogenation. Chem. Eng. J. 284, 260–269.

Kotas, T.J., 1995. The Exergy Method of Thermal Plant Analysis. Krieger Publishing Company, Malabar, Florida.

Larson, E.D., Consonni, S., Kreutz, T.G, 2000. Preliminary economics of black liquor gasifier/gas turbine cogeneration at pulp and paper mills. J. Eng. Gas Turbines Power 122, 255–261.

Leonzio, G., 2018. State of art and perspectives about the production of methanol, dimethyl ether and syngas by carbon dioxide hydrogenation. J. CO_2 Util. 27, 326–354.

Leonzio, G., 2020. Mathematical modeling of a methanol reactor by using different kinetic models. J. Ind. Eng. Chem. 85, 130–140.

Lücking, L.. 2017. Methanol Production from Syngas: process modelling and design utilising biomass gasification and integrating hydrogen supply. Available at: https://repository.tudelft.nl/islandora/object/uuid%3Ac0c5ebd2-c336-4f2d-85d1-014dae9fdf24.

Luyben, W.L., 2011. Principles and Case Studies of Simultaneous Design. John Wiley & Sons.

Macedo, J.C.F., 2006. Thermal and Environmental Analysis of the Primary Sludge from Pulp and Paper Making in Biomass Boiler On the Grid. MSc Disssertation, Federal University of Itajubá [In Portuguese].

Mevawala, C., Jiang, Y., Bhattacharyya, D., 2017. Plant-wide modeling and analysis of the shale gas to dimethyl ether (DME) process via direct and indirect synthesis routes. Appl. Energy 204, 163–180.

Monge, C. 2018. Colombia puts a tax on carbon [Online]. Available: https://www.conservationfinancenetwork.org/2018/11/27/colombia-puts-tax-on-carbon [Accessed].

Mongkhonsiri, G., Gani, R., Malakul, P., Assabumrungrat, S., 2018. Integration of the biorefinery concept for the development of sustainable processes for pulp and paper industry. Comput. Chem. Eng. 119, 70–84.

Moraes, F.D.A.B., Piratelli, Claudio Luis e Achcar, Jorge AlbertoCondições ideais para o consumo específico de madeira na produção de celulose. Production [online]. 2014, v. 24, n. 3 [Acessado 30 Abril 2022] , pp. 712-724. Disponível em: https://doi.org/10.1590/S0103-65132013005000074. Epub 24 Set 2013. ISSN 1980-5411.

Moraes, F.D.A.B., 2011. Model For the Evaluation of the Specific Consumption of Wood and Energy Input in the Process of Pulp and Paper Manufacturing. Araraquara University Center [In portuguese]. MSc dissertation. Available at: https://www.uniara.com.br/arquivos/file/ppg/engenharia-producao/producaointelectual/dissertacoes/2011/francisco-assis-bertini-moraes.pdf.

Naqvi, M., Yan, J., Dahlquist, E., 2010a. Black liquor gasification integrated in pulp and paper mills: a critical review. Bioresour. Technol. 101, 8001–8015.

Naqvi, M., Yan, J., Fröling, M., 2010b. Bio-refinery system of DME or CH4 production from black liquor gasification in pulp mills. Bioresour. Technol. 101, 937–944.

Nakashima, R., Florez-Orrego, D., Oliveira Junior, S. (2019) Integrated anaerobic digestion and gasification processes for upgrade of ethanol biorefinery residues. International Journal of Power Technologies. v. 99 No 2. pp. 104–114.

Nimkar, S.C., Mewada, R.K., Rosen, M.A., 2017. Exergy and exergoeconomic analyses of thermally coupled reactors for methanol synthesis. Int. J. Hydrog. Energy 42, 28113–28127.

Ott, J., Gronemann, V., Pontzen, F., Fiedler, E., Grossmann, G., Kersebohm, D.B., Weiss, G., Witte, C., 2012. Methanol. Ullmann's Encyclopedia of Industrial Chemistry. Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim, Germany.

Pettersson, K., Harvey, S., 2012. Comparison of black liquor gasification with other pulping biorefinery concepts–Systems analysis of economic performance and CO_2 emissions. Energy 37, 136–153.

Pio, D.T., Tarelho, L.A.C., Matos, M.A.A., 2017. Characteristics of the gas produced during biomass direct gasification in an autothermal pilot-scale bubbling fluidized bed reactor. Energy 120, 915–928.

Rural Technology Initiative. 2016. Chapter 8 . Pulp and Paper [Online]. Available: http://www.ruraltech.org/projects/conversions/briggs_conversions/briggs_ch08/chapter08_combined.pdf [Accessed 17 September 2018].

Szargut, J., Morris, D., Steward, F., 1988. Exergy Analysis of thermal, chemical, and Metallurgical Processes. Hemisphere Publishing Corporation, New York.

Theis, M., Skrifvars, B.J., Zevenhoven, M., Hupa, M., Tran, H., 2006. Fouling tendency of ash resulting from burning mixtures of biofuels. part 2: deposit chemistry. Fuel 85, 1992–2001.

Tountas, A.A., Peng, X., Tavasoli, A.V., Duchesne, P.N., Dingle, T.L., Dong, Y., Hurtado, L., Mohan, A., Sun, W., Ulmer, U., 2019. Towards solar methanol: past, present, and future. Adv. Sci. 6, 1801903.

Turton, R., Whiting, W.B., Bhattacharyya, D., Shaeiwitz, J.A.., 2018. Analysis, synthesis, and design of chemical processes. Prentice Hall, New York.

UNIDO. 1996. Evaluation of technologies on recovery of black liquor chemicals for small and medium size pulp mills using non-wood fibrous raw materials [Online]. United Nations Industrial Development Organization. Available: https://open.unido.org/api/documents/4808393/download/EVALUATION [Accessed 5 August 2018].

Vakkilainen, E.K. 2006. Kraft recovery boilers – high dry solids firing. 5–254. Available at: https://www.researchgate.net/profile/Esa-Vakkilainen/publication/279871546_Kraft_recovery_boilers_-_Principles_and_practice/links/561a6dba08ae6d1730898b68/Kraft-recovery-boilers-Principles-and-practice.pdf.

Van Rens, G., Huisman, G., De Lathouder, H., Cornelissen, R., 2011. Performance and exergy analysis of biomass-to-fuel plants producing methanol, dimethylether or hydrogen. Biomass Bioenergy 35, S145–S154.

Velásquez, H., De Oliveira, S., Benjumea, P., Pellegrini, L., 2013. Exergo-environmental evaluation of liquid biofuel production processes. Energy 54, 97–103.

Velásquez, H.I., Colorado, A.R., Júnior, S.D.O., 2009. Ethanol production from banana fruit and its lignocellulosic residues: exergy and renewability analysis. Int. J. Thermodyn. 12, 155–162.

WBG, 2020. State and Trends of Carbon Pricing 2020. World bank Group, Washington, DC Available at: https://openknowledge.worldbank.org/handle/10986/33809.

Williams, R.H., Larson, E.D., Katofsky, R.E., Chen, J., 1995. Methanol and hydrogen from biomass for transportation. Energy Sustain. Dev. 1, 18–34.

Yang, S., Li, B., Zheng, J., Kankala, R.K., 2018. Biomass-to-Methanol by dual-stage entrained flow gasification: design and techno-economic analysis based on system modeling. J. Clean. Prod. 205, 364–374.

Yang, Y., Mims, C.A., Mei, D., Peden, C.H., Campbell, C.T., 2013. Mechanistic studies of methanol synthesis over Cu from $CO/CO_2/H_2/H_2O$ mixtures: the source of C in methanol and the role of water. J. Catal. 298, 10–17.

Yoo, M., Lessard, L., Kermani, M., Maréchal, F., 2015. OSMOSE Lua: a unified approach to energy systems integration with life cycle assessment. In: Proceedings of the 12th International conference PSE 2015 and 25th International conference ESCAPE 2015, 2015 Copenhagen. Denmark.